



Regulatory transparency and the alignment of private and public enforcement: Evidence from the public disclosure of SEC comment letters[☆]

Amy Hutton^{a,*}, Susan Shu^a, Xin Zheng^b

^a Carroll School of Management, Boston College, United States

^b Sauder School of Business, University of British Columbia, Canada

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ABSTRACT

Does enhanced regulatory transparency facilitate alignment of private and public enforcement? Utilizing the SEC's 2004 decision to publicly disclose its comment letters, we explore the actions of the SEC and shareholder litigants. We find the two parties converge more on enforcement targets after the public disclosure. The increased alignment is attributable to public scrutiny of SEC oversight enhancing regulator incentives and reducing regulatory capture, and to shareholder plaintiffs gaining information previously accessible only by regulators, enabling litigants to identify cases with "merit." These findings suggest regulatory transparency enhances the complementarity of public and private enforcement, potentially improving enforcement outcomes.

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1. Introduction

The effectiveness of private and public enforcement mechanisms in constraining managerial misconduct has garnered widespread interest from regulators, practi-

tioners, and academics. Public enforcement involves government entities, such as the US Securities and Exchange Commission (SEC), while private enforcements are actions taken by private parties, such as shareholders. Regulators face resource constraints and can be captured by special interests (Stigler, 1971). Private parties, while generally apolitical, have limited enforcement options and information access. Thus, efficient enforcement often involves blended enforcement, with engagement from both private and public mechanisms (Djankov et al., 2003; Shleifer, 2005). In this paper, we propose enhanced regulatory transparency as a way to improve the complementarity between the public and private enforcers. Specifically, greater regulatory transparency facilitates the

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* Corresponding author.

E-mail addresses: amy.hutton@bc.edu (A. Hutton), susan.shu@bc.edu (S. Shu), xin.zheng@sauder.ubc.ca (X. Zheng).

monitoring of regulators and allows private parties access to information previously only privy to regulators. The alignment in incentives and information likely results in greater overlap in enforcement actions as the two mechanisms converge on higher-quality cases, potentially improving enforcement outcomes.¹

To test our conjecture, we focus on the interaction between the SEC, a public enforcer, and private shareholder litigants, a private enforcement mechanism, following an SEC policy change that initiated the public disclosure of SEC comment-letter correspondence. Specifically, in June 2004, the SEC announced a shift in policy: starting with corporate reports filed after August 1, 2004, the agency would publicly disclose its comment letters and corporate filers' responses. Before this policy change, comment letters were only accessible to parties who filed a Freedom of Information Act (FOIA) request. The stated objective of the policy change was to alleviate both the delay and the selective access to SEC comment letters. In making the comment letters public, the SEC increased the transparency of its oversight activities.² We examine whether this increased transparency facilitates the alignment of the public and private enforcers' incentives and information sets, resulting in a greater overlap of enforcement actions. Our evidence suggests that public enforcement (via SEC comment letters) and private enforcement (via private securities litigation) coincide to a greater extent after the SEC comment letters are released publicly. Additionally, there appears to be an increased alignment in timing: the (absolute) filing lag between private and public enforcement activities declines significantly following the SEC's disclosure policy change.

Having demonstrated greater alignment in enforcement following the public disclosure of SEC comment letters, we turn to the specific factors contributing to the alignment. First, we consider the SEC's incentive alignment: Did the SEC increase enforcement efforts after its enforcement activities became more observable and easily monitored? We find that the SEC steps up its oversight activities during the post-public disclosure period. Specifically, the SEC is more likely to issue comment letters to firms with questionable accounting practices (i.e., firms that eventually must restate their financials).

In addition to increased enforcement efforts, the SEC is also less susceptible to political influence in the post-

disclosure period. Whereas the SEC was less likely to enforce against politically connected firms in the pre-disclosure period when the SEC's oversight activities were not readily observable, we find a positive association in the post-disclosure period, consistent with Heese et al. (2017). The conflicting evidence in the two disclosure regimes provides a reconciliation of contradictory findings in Correia (2014) and Heese et al. (2017): While the SEC can be captured when its actions are opaque to the public, such incentives are mitigated, even reversed, when the SEC's actions become visible to the public. In sum, the SEC's increased tendency to issue comment letters to restating firms and to politically connected firms suggests that regulatory transparency reduces political capture and enhances regulators' incentives to detect financial reporting problems.

Next, we turn to the role of information alignment. We conjecture that public disclosure of SEC comment letters enhances private parties' information access by revealing information previously only privy to the SEC.³ The narrowed information gap between private litigants and the SEC likely results in a greater overlap of enforcement targets. We conduct two sets of tests of the information channel. First, we conduct cross-sectional tests that examine whether the increased overlap is greatest for firms with more information asymmetry (i.e., smaller, younger firms with less institutional ownership and fewer analysts following). Our findings are broadly consistent with these predictions.

In our second set of tests of the information channel, we examine whether the availability of information from SEC's filing reviews improves private litigants' ability to identify "quality" lawsuits. With access to SEC comment letter correspondence, plaintiffs' attorneys can more easily target firms with questionable financial reporting practices, thereby reducing nuisance lawsuits. We hypothesize that the increased merit of securities litigation translates to a lower likelihood of case dismissal and larger settlement amounts. In this analysis, we focus on the effect of enhanced information access on the merit of cases pursued by securities lawyers. To isolate the impact of the litigants' information access and avoid the confounding effects of the changing SEC's incentives across the two disclosure regimes, we focus exclusively on the post-disclosure period. To capture differential information access, we exploit the relative timing of a lawsuit filing date and the related comment letter dissemination date. If a lawsuit is filed after (before) the relevant comment letters are disseminated, then the plaintiffs have (do not have) access to relevant correspondence between the SEC and the registrant.

¹ Greater overlap demonstrates the complementarity of public and private enforcement. Private mechanisms pick up where the public enforcement stops (due to resource constraints or incentive problems). Aided by greater regulatory transparency, private enforcers can better target corporate wrongdoing and impose additional costs on the bad actors. Ultimately, the alignment of the two mechanisms leads to more efficient enforcement if the engagement of both holds more corporate wrongdoers accountable while reducing deadweight loss associated with nuisance cases.

² SEC oversight activities range from the Division of Corporate Finance (DCF)'s review and comment process to the Division of Enforcement's (DOE's) issuance of Accounting and Auditing Enforcement Releases (AAERs). The latter involves the most egregious fraud cases and captures only a small fraction of the SEC's overall enforcement. Most accounting and disclosure deficiencies that come to the SEC's attention are resolved through the SEC's review and comment process (Bayless, 2000; Heese et al., 2017).

³ Plaintiff lawyers could, in theory, request comment letter correspondence via FOIA before 2004. However, access to such correspondence was severely limited. First, plaintiff attorneys had no reliable means of identifying which firms received comment letters, so they could not file relevant FOIA requests. For the SEC to process a FOIA request, the request had to name a specific company for specified dates (i.e., one could not ask for, say, all comment letters issued in 2002 to Fortune 500 firms). Duro et al. (2019) highlight that before 2004, only 56% of the comment letter FOIA requests were "accepted" by the SEC, and the SEC took an average of 697 days to respond.

We find that access to comment letters indeed improves the merits of the shareholder litigation: lawsuits filed after the dissemination of the SEC comment letter correspondence are less likely to be dismissed, and they result in larger settlement amounts than those filed before the dissemination date.⁴ Overall, this evidence highlights that a transparent SEC review process lowers private litigants' information search costs and encourages more effective private enforcement of securities laws (Landis, 1938; La Porta et al., 2006).

Since the SEC's new disclosure policy went into effect for all registrants and all filings filed after August 1, 2004, we do not have a control sample for which comment letters are issued but not publicly disclosed in the post-period.⁵ That is, our setting does not lend itself to a difference-in-differences design. Thus, our inferences are susceptible to confounding factors and alternative explanations. We conduct a series of tests to solidify our inference that increased regulatory transparency is the most likely driver of increased public and private enforcement alignment. Importantly, our year-by-year regressions reveal that the increased alignment closely tracks the public releases of comment letters, as it occurs immediately after the transitional period surrounding the public disclosure with no evidence of similar effects in the prior period. In addition, we explicitly address various plausible confounding factors. First, we address the confounding impact of Section 408 of SOX by redefining the pre-period to include only the post-SOX sample. We find that the increased alignment between public and private enforcements became significant only after the public disclosure of comment letters. Next, the disclosure of internal control weaknesses (ICW) arising from SOX could also drive the increased alignment. When we include reported ICWs as an additional control or rerun our tests excluding firm-years with ICWs, the tenor of the results is unaffected. Other confounding factors include shifting political ideology of Congressional committee members overseeing the SEC and improved quality of plaintiff representation. Incorporating politicians' ideological leanings and plaintiff lawyers' quality leaves the inferences intact. Moreover, content analyses of SEC comment letters before and after the policy change suggest that changing content is unlikely to drive our results. To address the possibility of broader shifts within the SEC, we also conduct a falsification test by replacing SEC comment letters with SEC investigations, which remain undisclosed throughout the sample period. We do not find an increased alignment of the two enforcement mechanisms in the falsification test. While the possibility of other confounding events remains, this series of tests substantially limits the set of such explanations. Specifically, the remaining alter-

native explanations would need to have a shift in a key determinant that precisely coincides with the public dissemination of SEC comment letters.

To our knowledge, we are the first to empirically examine whether increased regulatory transparency affects the interaction between private shareholder litigation and the SEC's public enforcement activities. While prior studies present mixed findings on whether market-based enforcement is a compliment or substitute for public enforcement, we use a new setting to examine how regulatory transparency changed the incentive and information alignment between private and public enforcement.⁶

In the next section, we provide institutional background on the SEC's review process of registrants' filings, highlight related studies and develop our empirical hypotheses. Section 3 describes our data. In Section 4, we outline our research design and present our primary findings. Section 5 provides additional analyses useful for interpreting the primary findings; Section 6 presents robustness tests, and finally, Section 7 concludes.

2. Institutional background, related literature, and hypothesis development

2.1. The SEC's disclosure review process

SEC staff members in the Division of Corporation Finance provide issuers with comments on their corporate filings when the staff believes the filing should be improved. They also issue a comment letter when they deem a filing to be materially deficient or require further clarification. Robert A. Bayless (2000), former Chief Accountant, Division of Corporation Finance (DCF), highlights the important role of the comment letter process:

What is commonly overlooked... is that most financial reporting problems are handled by the Commission's full disclosure program, which is administered by the Division of Corporate Finance... Only the most egregious and obvious of these accounting errors lead to action by the Division of Enforcement. In the overwhelming majority of cases, the registrant restates its financial statements quietly after a challenge by the Division of Corporate Finance... [T]he accounting issues underlying the restatement of financial statements in these situations are usually no less important than those that have served as the basis for more dramatic enforcement actions.⁷

The DCF's disclosure reviews often involve several rounds of comments from the staff and responses from the

⁴ The larger settlement could also result from private plaintiffs using comment letters to boost their claims, "including materiality, scienter, loss causation, and more" (Platt, 2020).

⁵ Specifically, for the control sample, we need firms not subject to the disclosure rule change. Additionally, the assignment of treatment and control groups should be random. Even if we were to create a quasi-control sample that is insensitive to the CL rule change, say, through an event study around the announcement of the SEC rule change, these firms are by nature very different from our treatment firms. This selection bias will no doubt complicate any diff-in-diff inferences.

⁶ Our empirical analysis demonstrates that increased regulatory transparency improves the quality of the SEC review process and results in greater alignment of public and private enforcement. However, we do not attempt to determine whether increased transparency affects alignment directly or only indirectly through its effect on SEC quality. It could be either or both, and it is not obvious how one would empirically separate the direct and indirect effects. Nevertheless, we do not view distinguishing between the direct and indirect effects of transparency on alignment as important to our arguments or inferences.

⁷ Bayless (2000): <http://www.sec.gov/news/speech/spch394.htm>.

registrant.⁸ The SEC's disclosure review process is tied to its stated goal of ensuring that "investors have access to high-quality disclosure materials that are useful to investment decision making" (SEC 2011). SOX Section 408 mandates that the SEC review all public companies' disclosures at least once every three years.

Before the policy change, the SEC only released staff comment letters and issuer responses selectively to those who filed FOIA requests.⁹ In June 2004, the SEC announced a change in policy regarding the release of comment letters and filer responses:

In recent months, an increasing number of our comment letters and filer responses to them are being released publicly through the FOIA process, but only to those persons who make FOIA requests for them. We believe it is appropriate to expand the transparency of the comment process so that this information is available to a broader audience, free of charge... Public access to this correspondence will no longer require a FOIA request.¹⁰

The SEC announcement indicated that it would begin releasing comment letters and response letters relating to disclosure filings after August 1, 2004. The SEC later announced that the first batch of comment letters would be publicly released on May 12, 2005.¹¹

The SEC announcement also indicated that the SEC staff "may ask companies to represent in writing that they will not use the SEC's comment process as a defense in any securities-related litigation."¹² The SEC's decision to include such language in all initial reviews demonstrates two important points. First, the SEC recognizes the link between information provided, vis-à-vis its review process, and private securities litigation. Second, the SEC does not want its disclosure review process to undermine the legitimacy of any private shareholder enforcement action.

2.2. Related literature

While many studies in comparative economics discuss various mechanisms for public control over economic activity, Segal and Whinston (2006) point out two basic ap-

proaches for deterring socially harmful behavior in most countries: private litigation and public enforcement by agencies such as the SEC. Empirical studies employ international settings to assess which control mechanisms are more beneficial to the capital market, with mixed findings. La Porta et al. (2006) examine the effect of securities laws on stock market development in 49 countries and find little evidence that public enforcement benefits stock markets. Instead, they find that stock markets benefit from laws that mandate disclosure and facilitate private enforcement through liability statutes. In contrast, Jackson and Roe (2009) employ securities regulators' resources as a proxy for regulatory intensity and find public enforcement to be as important as disclosure and more important than private liability standards in explaining financial market outcomes.

Private enforcement of public statutes is a highly effective strategy of enforcing good conduct in many situations (Hay et al., 1996; Hay and Shleifer, 1998), empirically demonstrated for securities markets by La Porta et al. (2006) and for banking regulation by Barth et al. (2004). Private securities litigation by shareholders is an example of private enforcement of corporate conduct under the Securities Acts of 1933 and 1934, providing an important means for shareholders to redress damages from managerial misconduct (Schantl and Wagenhofer, 2020). The threat of shareholder lawsuits affects various corporate decisions (e.g., Skinner 1997, Field et al. 2005 and Hopkins 2018). However, legal scholars have also questioned the effectiveness of private securities litigation. Plaintiff attorneys' incentives to go after deep pockets and to race to the courtroom, combined with their information disadvantage, lead to nuisance lawsuits that cause deadweight welfare loss (e.g., Rose 2008 and Mahoney 2009).

Some studies highlight several distinct advantages of a public enforcer, such as the SEC, when private enforcement incentives are insufficient (La Porta et al., 2006). Public regulators can develop expertise and be motivated to pursue social objectives (Landis, 1938; Glaeser et al., 2001; Pistor and Xu, 2002). They have a greater array of enforcement options and enjoy exclusive access to non-public information they can solicit directly from firms. These various advantages make public enforcement in some circumstances more efficacious than private enforcement (Shleifer, 2005). Indeed, SEC oversight has been shown to deter managerial misconduct (Jackson and Roe, 2009; Kedia and Rajgopal, 2011). However, many argue that the SEC is susceptible to political influences (e.g., Shleifer 2005, Correia 2014, Mehta and Zhao 2020, Zheng 2021 and Yu and Yu 2011), undermining its enforcement integrity. In addition, government bureaucracy can expend insufficient effort to detect financial reporting problems (Prendergast, 2003, 2007; Schantl and Wagenhofer, 2020). Although both the SEC and the securities litigation target corporate wrongdoings, each exhibits biases and inefficiencies.

With public access to SEC comment letters, several recent papers have begun to explore their information and financial reporting consequences. Johnston and Petacchi (2017) find that following comment letter resolution,

⁸ As noted by Bayless in the above quote, the DCF infrequently recommends the company and issues discussed in its CLs to the Enforcement Division. Rather, our understanding is that a large majority of the Division of Enforcement's activities begin with an anonymous tip. According to the SEC's annual enforcement report, "Commission personnel reviewed more than 16,000 tips, largely from the general public." (SEC, 2017)

⁹ Given the statute of limitations on 10(b)–5 claims is two years after fraud discovery and no more than five years after fraud occurrence (Citation: 28 U.S.C. § 1658(b)), the lengthy delay in the SEC's response to FOIA requests largely rendered CL correspondence useless to plaintiffs before 2004.

¹⁰ See SEC press release: <https://www.sec.gov/news/press/2004-89.htm>.

¹¹ While the SEC indicated that CL correspondence would be released 45 days after the staff completed its filing review, we identify a transition period for the SEC's CL disclosure policy from August 2004 to June 2006, during which time disclosure of CL correspondence was substantially delayed. See the detailed discussion of the rollout of the SEC's disclosure of CLs in Section 3.2.

¹² This request is known as a "Tandy" letter. Retrieved from <https://www.sec.gov/news/press/2004-89.htm>. The SEC discontinued the Tandy letter language in October 2016.

the adverse selection component of the bid-ask spread declines, and earnings response coefficients increase, concluding that the SEC's oversight has beneficial informational effects. [Duro et al. \(2019\)](#) find evidence that after the public disclosure of comment letters, targeted firms' financial reporting improves.¹³

2.3. Hypotheses development

While there are many studies that examine the SEC and shareholder litigants' oversight and enforcement activities, our understanding of how these two enforcement mechanisms interact is limited. Two exceptions are [La Porta et al. \(2006\)](#) and [Jackson and Roe \(2009\)](#). Both use international settings to explore the importance of public and private enforcement in capital markets development at the country level. In this paper, we investigate this interaction by examining whether the increased transparency of the SEC oversight and enforcement process affects its behavior and the outcomes of private enforcement.

We expect the 2004 policy change to enhance the SEC's effectiveness as a public enforcer. According to economic theory, public enforcers are effective if they are independent (not politicized and not captured), motivated to pursue social objectives, and if they can secure information from issuers that private parties cannot (or at least not as efficiently). Making the SEC enforcement activities transparent enhances the external monitoring of the SEC, which in turn, should improve the SEC's incentives to be independent (or at least appear independent). Similarly, greater transparency likely motivates the SEC's staff to step up enforcement activities important to investors (e.g., high profile cases that result in restatements).¹⁴ Moreover, after the SEC policy change, any new information the SEC secured from issuers via the comment letter review process is revealed to outsiders, ensuring greater information alignment across private and public enforcement.¹⁵ The SEC's incentive alignment and shareholder litigants' information

alignment after the public disclosure should result in both enforcement mechanisms targeting firms with questionable accounting and disclosure practices. This leads to our formal hypotheses:

H1: After the public disclosure of SEC comment letters, there is an increased alignment in enforcement activities between the SEC (via comment letters) and private shareholder litigants.

H2a (Incentive Alignment): The SEC is more likely to issue comment letters to firms with questionable accounting practices.

H2b (Incentive Alignment): The SEC is less likely to be captured by politically connected firms.

H3 (Information Alignment): The increased alignment in enforcement activities is more pronounced in firms with greater information asymmetry.

H4 (Information Alignment): Private securities lawsuits filed with access to SEC comment letters have more merit (i.e., experience lower dismissal rates and higher settlement amounts than lawsuits filed without such access).

3. Data sources and the rollout of public disclosure of SEC comment letters

3.1. Data and sample selection

We begin with the Compustat database to identify all firm years in our sample period, 1997 to 2015. We then augment Compustat with data on SEC comment letters (CL) and shareholder class action lawsuits. The CL sample comprises all CLs issued for 10-Ks filed between January 1, 1997 and June 30, 2015. [Dechow et al. \(2016\)](#) show that SEC CLs predominantly relate to annual reports. Each CL refers to one 10-K filing (i.e., alleged wrongdoing year). Since the SEC began to publicly disclose all CLs related to filings filed after August 1, 2004, we gather the post-2004 CLs from Audit Analytics. For the pre-2004 CL reviews, we obtain data through FOIA requests. The SEC provided the recipients of the CLs (company names) and the first and last correspondence dates.¹⁶

We next search Stanford's Securities Class Action Clearinghouse for all shareholder lawsuits with an alleged Rule 10b-5 violations filed within our sample period, January 1, 1997 through June 30, 2015, and hand-collect detailed lawsuit information (<http://securities.stanford.edu/index.html>). We focus on lawsuits against public companies and exclude cases related to IPO allocations, mutual funds, and analysts, following prior studies (e.g., [Kim and Skinner 2012](#)). Our sample period starts after the Private Securities Litigation Reform Act (PSLRA, 1996), which ensures consis-

¹³ Interestingly, [Duro et al. \(2019\)](#) do not find evidence of increased SEC oversight intensity after 2004, as measured by the length of the CL review process or the number of rounds of letters. In addition, the authors analyze the determinants of receiving a CL for the periods before and after the SEC disclosure policy change. Based on the low explanatory power of the determinant models both before and after the disclosure policy change, the authors conclude that "by and large, firms are systematically subject to CL reviews."

¹⁴ Section 408 (b1) of SOX outlines criteria for when the SEC should review an issuer's filings. These criteria include restatements, suggesting that SOX likely led to an increase in the overlap of reviews and restatement firms. However, a filing review per se does not necessarily indicate that a comment letter will be sent to the issuer. Nevertheless, in our empirical tests, we use sub-sample analyses to estimate the effects of SOX and the SEC's decision to disclose all comment letter correspondence separately.

¹⁵ [La Porta et al. \(2006\)](#) argue that to effectively harness market participants' incentives to enforce securities laws through private litigation, it is necessary to make the private recovery of investors' losses easy ([Landis, 1938](#)). Such efficiency considerations include lower information and search costs for market participants through disclosure requirements and liability standards that make it cheaper for investors to recover damages when information is wrong or omitted. We add to these arguments the insight that the SEC's review process, when made public, brings to light new information that lowers private litigants' information and search costs.

¹⁶ For the pre-2004 period, we do not have access to the text of the CLs. Nor do we have access to which CLs were disclosed to interested parties in response to a FOIA request. (The SEC's FOIA log is available only post 2006: <https://www.sec.gov/foia/docs/foia-logs.htm>) Thus, while we confirm the alignment of disclosure and accounting topics of the CL and private litigation actions in the post-2004 period, we assume this alignment holds in the pre-2004 period. Additionally, we assume that SEC CLs were not disclosed to private litigants in the pre-2004 period, although they might have been via a FOIA request. We note that a violation of these assumptions would weaken the power of our tests but do not necessarily introduce a bias into our analysis.

tent legal standards throughout our sample period. It ends in June 2015 to allow sufficient time for lawsuit resolution.¹⁷

We first identify firm-year observations whose 10-Ks are associated with SEC CLs. We then identify all firm years associated with securities litigation.¹⁸ To determine the firm years subject to both public oversight and private action, we require an overlap between the lawsuit class period and the CL subject year (from the start of a 10-K fiscal year to its filing date). This matching criteria likely identifies the CLs and class action lawsuits that target the same wrongdoing years.¹⁹

Our control variables are motivated by two separate lines of research. First, we include control variables related to the securities litigation literature (e.g., Kim and Skinner 2012). We control for high litigation industries (*Litigation_ind*), aggregate abnormal returns in the year before the 10-K period (*Returns*), stock turnover (*Turnover*), and stock skewness (*Skewness*). Second, we include control variables related to the issuance of SEC CLs (e.g., Cassell et al. 2013, Dechow et al. 2016 and Heese et al. 2017). Specifically, we control for high firm-level stock price volatility (*High_volatility*), firm's market capitalization (*Size*), market-to-book ratio (*M2B*), firm age (*Age*), loss firms (*Loss*), Altman Z-score (*Altman*), changes in sales (*Chg_Sales*), merger activities (*Merger*), restructuring costs (*Restructure*), restatements in the current or prior years' 10-Ks (*Rstmt* (*t* or *t-1*)), amendments to the firm's 10-Ks (*Amend*), natural log of the number of days between fiscal year-end date and 10 K filing date (*FilingGap*), financing activities (*Financing*), Big 4 auditors (*Big4*), and auditor tenure (*Tenure*). We use the Government Accountability Office (GAO) restatement data and Audit Analytics to identify restatements in the pre- and post-period, respectively. We acquire political action committee (PAC) contributions from the Center for Responsive Politics (<https://www.opensecrets.org/>). Variable definitions are in the Appendix.

¹⁷ On average, the median time to settle a class action lawsuit is 3.2 years (retrieved from <https://corpgov.law.harvard.edu/2020/03/11/securities-class-action-settlements-2019-review-and-analysis/>).

¹⁸ In cases involving misleading statements or omissions, a class period generally starts when a company allegedly makes an untrue statement or omits material facts and ends with the revelation of the "truth" to the investing public via a "corrective disclosure." On average, 2.3 firm years fall into a class period for our sample lawsuits. As a result, a lawsuit can be matched with more than one 10-K that received a CL. Thus, we clustered standard errors by firm. We cannot cluster standard errors by lawsuits, as most of our firm-year observations are not associated with a case.

¹⁹ To further validate that the matched CLs and legal complaints indeed focus on the same disclosure and accounting issue(s), we examined all firm years with both a CL and a class action lawsuit in the post-public disclosure period. We started with issues identified in the CLs. The SEC staff listed all the issues in bullet points in CLs. We then compared these with issues identified in class action complaints. For over 75% of the observations, the financial reporting issues outlined in the CL and complaint overlap. As an example: 3D Systems Corporation received a CL on its 2013 10-K. The focus of the SEC's CL review (from May 13, 2014 to June 5, 2014) was the company's method for calculating its organic growth rate and questions regarding its disclosure of revenues from acquired businesses for each period presented. The class action lawsuit filed on June 12, 2015 highlighted several corrective disclosures during the class period (October 29, 2013 to May 5, 2015) that also focused on the company's reported (and lack of reporting of) "organic growth rates."

We describe the sample selection procedure in Panel A of Table 1. The final sample consists of 25,267 firm-years in the pre-disclosure period and 28,625 firm-years in the post-disclosure period. To assess the impact of SOX, we further split the pre-disclosure period into pre- and post-SOX with 18,911 and 6,356 firm-year observations, respectively. While in the pre-disclosure period, 11.9% of the firm-years receive CLs, it is worth noting that the percentage is lower pre-SOX, 8.9%, and increases to 20.7% with the introduction of SOX's Section 408 requirement that the SEC review all issuers' 10-Ks at least once every three years. Interestingly, the rate of firm-years receiving CLs rises to 32.2% in the post-disclosure period. Examining the fraction of firm-years associated with lawsuits, it is similar between the pre-SOX period (5.9%) and the post-disclosure period (5.3%). Notably, it spikes to 7.6% in the post-SOX, pre-disclosure period (between 2002 and 2004). The question of interest is whether the overlap between SEC CLs and private class action lawsuits increases in the post-2004 period. In fact, the frequency of firm-years targeted by both shareholder litigants and the SEC appears to be quite similar in the pre-disclosure but post-SOX period (2.3%) and the post-disclosure period (2.5%). Below, we formally test whether there is a change in the frequency of the overlap in a multivariate setting.

In Panel B of Table 1, we provide descriptive statistics on the CL review process pre- and post-2004. The filing lag (the number of days between the 10-K filing and the first CL) appears similar pre- and post-2004. However, the number of days between the start and finish of the review declined in the post-period. The mean number of days for a review fell from 114 days to 71 days (median fell from 72 days to 49 days). This decline could result either from changes in SEC staff behavior or changes in issuers' behavior (Duro et al., 2019). Finally, for completeness, Panel C of Table 1 provides descriptive statistics for the control variables.

3.2. Transition period: August 2004 to June 2006

While the SEC announced in June 2004 that the policy change would apply to all filings after August 1, 2004, it is important to note that the SEC's new CL disclosure policy experienced a transition period. Specifically, due to a variety of reasons including technical hurdles, public dissemination of CLs was sparse until the second half of 2006, when the SEC publicly released a large batch of CLs.²⁰

To better understand the transition period of the SEC's new disclosure policy, we plot the CLs' dissemination fre-

²⁰ Director of the SEC's DCF, John W. White, in a May 28, 2006 NYT article suggested that technical challenges were to blame for the untimely disclosures of CLs between August 2004 and June 2006: "We have now resolved the technical hurdles of posting the information on Edgar, and we have committed our resources to getting correspondence arising since August 2004 posted as soon as possible so that it will be readily available to all investors free of charge. We expect a significant number of new posting in the coming months" (Morgenson, 2006).

Table 1
Descriptive statistics.

		Pre-period before SOX Jan 1, 1997 - July 31, 2002	Pre-2004 Pre-period after SOX Aug 1, 2002 - July 31, 2004	Combined Jan 1, 1997 to July 31, 2004	Post-2004 (Aug 1, 2004 to June 30, 2015)
Panel A: Firm-year sample selection					
Firm years in final sample		18,911	6,356	25,267	28,625
Firm years with CL (%)		1,696 (8.9%)	1,317 (20.7%)	3,013 (11.9%)	9,208 (32.2%)
Firm years with lawsuits (%)		1,134 (5.9%)	487 (7.6%)	1,621 (6.4%)	1,509 (5.3%)
Firm years with CL and lawsuits (%)		175 (0.9%)	145 (2.3%)	320 (1.2%)	704 (2.5%)
Panel B: The SEC's filing review process (SEC comment letters)					
CL Review Period (in days)	N	1,696	1,317	3,013	9,208
	Mean	125	98	114	71
	Median	77	66	72	49
CL Filing Lag (in days)	Mean	162	128	148	149
	Median	153	111	133	135
Panel C: Summary statistics for control variables for the 53,892 firm-years used in our final sample					
	25%tile	Median	75%tile	Mean	SD
<i>High_volatility</i>	0	0	0	0.105	0.307
<i>Size</i>	4.156	5.736	7.240	5.766	2.137
<i>M2B</i>	1.217	2.022	3.542	3.272	4.255
<i>Age</i>	9	15	29	20.563	15.280
<i>Loss</i>	0	0	1	0.335	0.472
<i>Altman</i>	1.609	3.072	5.212	4.181	6.190
<i>Chg_Sales</i>	-0.028	0.077	0.223	0.172	0.527
<i>Merger</i>	0	0	0	0.102	0.302
<i>Restructure</i>	0	0	0	0.218	0.413
<i>Financing</i>	-0.049	-0.002	0.054	0.031	0.175
<i>Litigation_ind</i>	0	0	1	0.332	0.471
<i>Big4</i>	0	1	1	0.722	0.448
<i>Tenure</i>	4	7	13	9.523	7.944
<i>Returns</i>	-0.314	-0.056	0.235	0.044	0.608
<i>Turnover</i>	11.278	12.803	14.107	12.682	2.010
<i>Skewness</i>	-0.207	0.301	0.867	0.353	0.848
<i>Rstmt (t or t-1)</i>	0	0	0	0.103	0.305
<i>Amend</i>	0.000	0.000	0.000	0.089	0.284
<i>FilingGap</i>	4.143	4.344	4.500	4.345	0.303

quency and lag in our sample. In Fig. 1a, we plot the number of CLs publicly disseminated by month. In Fig. 1b, we plot the average dissemination lag for CLs released each month. The dissemination lag is the number of days from the last CL to the day the CL is posted on EDGAR. These figures demonstrate two important facts. First, a meaningful number of CLs is not disseminated until the second half of 2006. Second, those disclosed before the third quarter of 2006 were stale (average dissemination lag of 307 days). Thus, August 2004 to June 2006 can be viewed as a transition period for the SEC's new disclosure policy. We plot the timeline of our sample period in Fig. 1c.

4. Research design and results

4.1. Alignment of private and public enforcement

We use the following logistic regression to explore whether the likelihood of a parallel shareholder lawsuit in concert with the SEC action (targeting the same firm years)

increases after the public disclosure of SEC CLs:²¹

$$Prob(Lawsuit) = f(CL, CL*Post, controls). \quad (1)$$

The dependent variable, *Lawsuit*, equals one if a portion of the firm year falls within the class period of a securities class action and is zero otherwise. *CL* equals one if the firm year received a CL on its 10-K from the SEC and is zero otherwise. *Post* is an indicator variable for the public disclosure period (i.e., firm years with 10-K filing dates after August 1, 2004). Given the transition period for the SEC's new disclosure policy, defining *Post* using August 1, 2004 (rather than July 1, 2006) reduces the power of our tests. Thus, when appropriate, we also rerun our tests excluding the rollout period. Observations are at the firm-year level.

²¹ We use logistic regressions due to the classic limitations of linear models with a dichotomous dependent variable: heteroskedasticity, non-normally distributed error terms, and non-linearity. The first two problems affect the standard errors, and the last one affects the plausibility of the coefficient estimates (Green, 2012). Importantly, the non-linearity problem is exacerbated when the true probability is close to zero or one (Hoffmann, 2016). Since the probability of a lawsuit is low (5%), the logistic model is more appropriate in our setting because it fits the data better than a linear model.

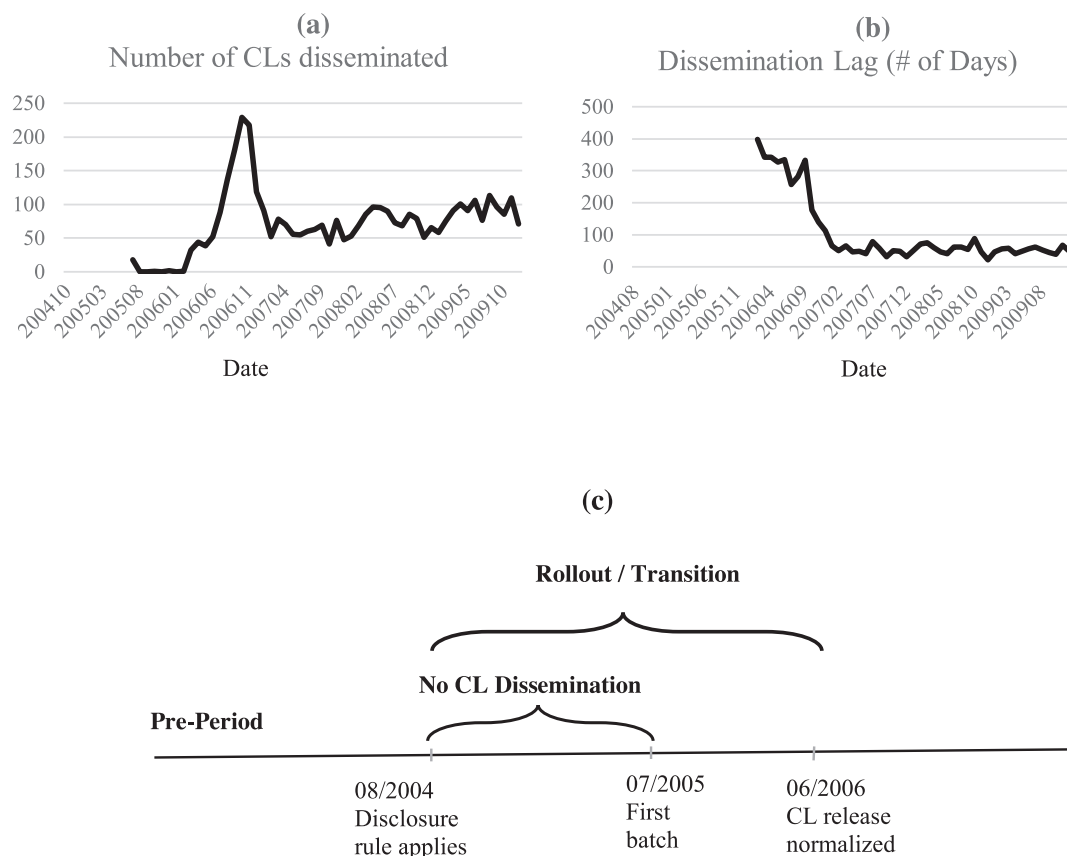


Fig. 1. Transitional period and timeline of CL disclosure rollout. Fig. 1 depicts the details of the rollout/transition period surrounding the 2004 SEC new policy change. Fig. 1(a) shows the number of CLs publicly disseminated by month. Fig. 1(b) shows the average dissemination lag for CLs released each month. The dissemination lag is the number of days from the last CL (closing letter) to the day the CL is posted on EDGAR. We illustrate the timeline of our sample period in Fig. 1(c).

We include industry (based on Fama and French's 12 industries) and year fixed effects or SEC office-year fixed effects to control for idiosyncratic industry or office and time effects. Standard errors are clustered at the firm level to adjust for time-series dependence (Petersen, 2009).

Table 2 presents the regression results. We estimate the relation between lawsuits and CLs for the pre-public disclosure period (column 1), the post-disclosure period (column 2), and the pooled regression with the interaction term $CL*Post$ (columns 3 through 6). The pooled regressions have different fixed effects. The regression for column 3 includes industry and year fixed effects, while the regression for column 4 includes SEC office-year fixed effects to control for idiosyncratic office and time effects. The regressions for columns 5 and 6 replicate the regressions presented in columns 3 and 4 but exclude observations from the rollout period (i.e., August 2004–June 2006). The findings consistently demonstrate a statistically significant increase in the association between SEC CLs and private lawsuits from the pre-2004 period to the post-2004 period. While the CL coefficient is positive and significant in the pre-disclosure period, it is significantly higher in the post-disclosure period. The chi-square test comparing the two CL coefficients rejects the equivalence of the coefficients

in the pre-period and the post-period. The coefficient on $CL*Post$ in columns 3 through 6 is positive and significant, confirming the stronger association between lawsuits and CLs in the post-public disclosure period. We further examine the odds ratio to assess the economic significance. Specifically, the odds ratios for column 3 (untabulated) suggest that in the pre-disclosure period, the odds of a lawsuit are 26% higher for firms with a CL than firms without a CL; in the post-period, they increase to 64% higher.²² Thus, the alignment or overlap of SEC CLs and private lawsuits increases significantly in the public disclosure period.²³

²² Odds ratios are the easiest ways to assess the economic significance of the coefficients in a logistic model with interactions and fixed effects. The odds ratio (untabulated) is 1.26 for CL and 1.3 for $CL*Post$. Thus, the odds ratio for a CL firm in the post period is 1.64 ($1.26*1.3$).

²³ The analysis for Table 2 includes many control variables that potentially affect litigation and CLs (Kim and Skinner, 2012; Cassell et al., 2013; Heese et al., 2017). One concern is that the findings could be driven by the stringent data requirement imposed by the control variable (which substantially reduces the sample size). To address this concern, we impose only minimal data requirements (firm-year observations have to have valid values for market capitalization and returns to ensure the observations are covered by Compustat and the Center for Research in Se-

Table 2

Alignment of lawsuits and comment letters.

	(1)	(2)	(3)	(4)	(5)	(6)
	Pre-2004	Post-2004	Pooled Yr FE	Pooled Office-Yr FE	Pooled Yr FE Excluding Rollout	Pooled Office-Yr FE Excluding Rollout
<i>CL*Post</i>			0.259*** (0.099)	0.258*** (0.100)	0.345*** (0.111)	0.346*** (0.112)
<i>CL</i>	0.181** (0.081)	0.564*** (0.072)	0.229*** (0.079)	0.228*** (0.079)	0.219*** (0.079)	0.218*** (0.080)
<i>High_volatility</i>	0.195* (0.108)	0.671*** (0.104)	0.474*** (0.074)	0.470*** (0.074)	0.480*** (0.079)	0.476*** (0.079)
<i>Size</i>	0.026 (0.039)	−0.005 (0.047)	0.011 (0.030)	0.004 (0.030)	0.014 (0.030)	0.006 (0.030)
<i>M2B</i>	−0.003 (0.008)	0.007 (0.008)	0.001 (0.005)	0.001 (0.005)	0.002 (0.006)	0.003 (0.006)
<i>Age</i>	−0.008* (0.004)	−0.013*** (0.004)	−0.010*** (0.003)	−0.012*** (0.003)	−0.010*** (0.003)	−0.012*** (0.003)
<i>Loss</i>	−0.031 (0.086)	−0.022 (0.090)	−0.018 (0.062)	0.014 (0.062)	−0.011 (0.064)	0.023 (0.064)
<i>Altman</i>	0.001 (0.005)	0.039*** (0.006)	0.017*** (0.004)	0.019*** (0.004)	0.015*** (0.004)	0.018*** (0.004)
<i>Chg_Sales</i>	0.206*** (0.038)	0.123*** (0.046)	0.170*** (0.030)	0.173*** (0.030)	0.180*** (0.031)	0.184*** (0.031)
<i>Merger</i>	0.231 (0.188)	0.110 (0.098)	0.076 (0.090)	0.082 (0.091)	0.063 (0.094)	0.071 (0.095)
<i>Restructure</i>	−0.003 (0.098)	−0.044 (0.085)	−0.046 (0.066)	−0.036 (0.066)	−0.067 (0.070)	−0.061 (0.070)
<i>Financing</i>	0.686*** (0.160)	−0.219 (0.195)	0.268** (0.128)	0.256** (0.129)	0.289** (0.133)	0.278** (0.135)
<i>Litigation_ind</i>	−0.253* (0.135)	−0.276** (0.137)	−0.241** (0.099)	−0.225** (0.099)	−0.263*** (0.100)	−0.245** (0.100)
<i>Big4</i>	−0.114 (0.098)	−0.283** (0.124)	−0.250*** (0.077)	−0.205*** (0.077)	−0.231*** (0.079)	−0.184** (0.080)
<i>Tenure</i>	−0.019*** (0.007)	−0.012 (0.008)	−0.017*** (0.005)	−0.017*** (0.006)	−0.019*** (0.006)	−0.019*** (0.006)
<i>Returns</i>	−0.133*** (0.048)	−0.500*** (0.083)	−0.260*** (0.042)	−0.261*** (0.043)	−0.239*** (0.043)	−0.242*** (0.044)
<i>Turnover</i>	0.544*** (0.045)	0.592*** (0.053)	0.548*** (0.034)	0.575*** (0.035)	0.538*** (0.035)	0.566*** (0.035)
<i>Skewness</i>	−0.257*** (0.042)	−0.322*** (0.039)	−0.304*** (0.028)	−0.300*** (0.029)	−0.300*** (0.030)	−0.296*** (0.030)
<i>Rstmt (t or t-1)</i>	1.595*** (0.100)	0.684*** (0.108)	1.157*** (0.075)	1.171*** (0.075)	1.201*** (0.076)	1.215*** (0.076)
<i>Amend</i>	0.154* (0.087)	0.410*** (0.092)	0.277*** (0.064)	0.265*** (0.065)	0.317*** (0.069)	0.308*** (0.069)
<i>FilingGap</i>	0.518*** (0.085)	1.522*** (0.135)	0.882*** (0.074)	0.897*** (0.074)	0.849*** (0.076)	0.866*** (0.076)
<i>Constant</i>	−13.081*** (0.649)	−17.129*** (0.857)	−14.688*** (0.569)	−14.616*** (0.911)	−14.465*** (0.575)	−14.419*** (0.916)
Observations	25,267	28,625	53,892	53,605	48,055	47,746
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No	Yes	No
SEC Office-Year FE	No	No	No	Yes	No	Yes
Pseudo R ²	0.204	0.168	0.172	0.183	0.173	0.184
Chi-square test of CL coefficients in (1) and (2)	χ^2 statistic: 13.46 P-value: <0.001					

This table reports the results of logistic regression $\Pr(\text{Lawsuit}) = f(\text{CL*Post}, \text{CL}, \text{controls})$. *CL* is an indicator variable equal to 1 if the 10 K filing in year *t* received a comment letter, and 0 otherwise. *Post* is an indicator variable equal to 1 if the time period is on or after August 1, 2004 and before June 30, 2015 and 0 otherwise. All other variables are defined in the Appendix. Column (1) uses the sample from pre-disclosure period; Column (2) uses the sample from post-disclosure period; Columns (3) and (4) include the pooled sample from pre- and post-periods with year fixed effects and SEC office-year fixed effects, respectively; Columns (5) and (6) exclude the rollout period from August 1, 2004 to June 30, 2006 and include the pooled sample from pre- and post-periods with year fixed effects and SEC office-year fixed effects, respectively. When including the SEC office-year fixed effects, the sample size decreases slightly as some office years are perfectly co-linear with the dependent variable. We report logistic coefficients in the table. Continuous variables are winsorized at 1% and 99%. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

As noted above, SOX (Section 408) required that the SEC review each issuer's filings (including 10-Ks) at least once every three years, likely contributing to the SEC's post-disclosure CL increase. However, of interest to us is whether more parallel private class action lawsuits accompany the SEC CLs in the post-2004 period. To disentangle the effect related to SOX, passed in 2002, we rerun the Table 2 regressions with the disclosure period split into two sub-periods: (1) Pre-SOX (January 1, 1997 through July 31, 2002, labeled *Pre-period 1*) and (2) Post-SOX but pre-disclosure (August 1, 2002 through July 31, 2004, labeled *Pre-period 2*). Table 3 follows the format of Table 2, except for the additional split for the two pre-periods, which are presented in columns 1 and 2, respectively. Columns 3 through 6 present the pooled regression results using only the second pre-period (i.e., Post SOX). Importantly, the coefficient on the interaction term $CL*Post$ is positive and significant in columns 3 through 6, suggesting the increased alignment in the post-disclosure period is not driven by the SOX effect.

Moreover, the sub-period analysis is instructive about the time series changes in the relation between lawsuits and CLs. Specifically, the coefficient estimate on CL in the second pre-period (post-SOX but pre-disclosure) is indistinguishable from that in the first pre-period (pre-SOX), as confirmed by the chi-square test results. In contrast, when comparing the coefficient on CL in the second pre-period (0.253) to that in the post-disclosure period (column 2 of Table 2, 0.564), the coefficient is significantly higher in the post-disclosure period (chi-square is 4.97, untabulated). These results are confirmed by the pooled regression results in columns 3 through 6.²⁴ Thus, while SOX could be responsible for the increase in the number of CLs issued after 2002, the increased alignment between CLs and securities class actions is unlikely to be driven by SOX. Instead, the increase in the parallel actions takes place only after the public disclosure of the CL correspondence.

To lend further support to our hypotheses, we estimate the year-by-year correlation between litigation and CLs. Specifically, we calculate by-year regression coefficient estimates of litigation on CLs and plot the coefficient estimates. If there is a clean break in the coefficient estimates consistent with the public dissemination of CLs, it will further solidify our inferences.

Fig. 2 presents the year-by-year regression coefficient estimates. We group observations into 12-month intervals to run the year-by-year regressions. As discussed earlier, the SEC's policy change experienced a transition period for various reasons, including technical difficulties that significantly delayed the public release of CLs. This transition is notable in the first 12 months of public dissemination

(July 2005 to June 2006), when the SEC released only 189 CLs with an average dissemination lag of 328 days. For comparison, during the next 12 months (July 2006 to June 2007), 1374 CLs are released with an average dissemination lag of 166 days. While we can estimate a coefficient for the transitional period, it is not surprising that the coefficient is statistically insignificant. Following the transition period (when the dissemination of CLs normalizes), the coefficient estimate immediately becomes positive and significant, as illustrated by the clear jump in the correlation between CLs and securities litigation coefficient in Fig. 2. Notably, the coefficient is positive and significant in each post-transition, year-by-year regression. This pattern suggests the robust nature of this relation and limits the number of possible alternative explanations. Nonetheless, to further corroborate the inferences of our main findings, we perform a series of falsification and robustness tests that address other plausible alternative interpretations (presented in Sections 5 and 6).

4.2. Incentive alignment: changes in SEC's behavior

Our evidence indicates an increased alignment in the enforcement targets of the SEC and private securities litigants after 2004. We explore two reasons for the increased alignment: The SEC's improved incentives and private litigants' enhanced information access. In this section, we focus on the SEC's behavior changes. With the increased transparency of its filing review process, we expect the SEC to become more vigilant in its enforcement activities and less captured by special interests in the post-disclosure period. Specifically, we test whether the SEC is more likely to issue CLs to firms that eventually restate their accounts and firms with significant PAC contributions in the post-2004 period.

We use the following logistic regressions to study the likelihood that the SEC issues CLs to firms with questionable accounting practices:

$$Prob(CL) = f(Rstmt, Rstmt*Post, controls). \quad (2)$$

The dependent variable, CL , equals one if the firm-year received a CL on its 10-K from the SEC and is zero otherwise. $Rstmt$ equals one if the firm's current or prior year's financials are eventually restated. Note a restated year refers to a wrongdoing year and not the year of the public announcement of the restatement. Existing evidence indicates that it takes one to three years for firms to announce the restatement of prior earnings (e.g., Dechow et al. 1996). In our restatement sample, the average lag from the end of the wrongdoing period to the restatement announcement date is almost two years (699 days). Thus, issuing a CL in the year after or the year of the financial misreporting effectively requires the SEC to spot these misstatements and raise concerns before the firm publicly announces the accounting irregularity. Another possibility is that the SEC CL process triggered the restatements, in which case the SEC's oversight role is even more salient. If, in the post-2004 period, the SEC is more likely to issue CLs to firms that eventually restate their financial statements, we expect the coefficient on $Rstmt * Post$ to be positive.

curity Prices (CRSP). We estimate a logistic regression of *Lawsuits* on only CL and $CL*Post$ using a sample of 77,522 firm years. The coefficient estimate on $CL*Post$ (0.196) is positive and highly significant, consistent with Table 2 results. Moreover, the coefficient on $CL*POST$ remains significant when we run the same analysis without controls using the final sample ($n = 53,892$).

²⁴ In terms of economic significance, the odds ratios for column 3 (untabulated) suggest that in the pre-disclosure but post-SOX period, the odds of a lawsuit are 22% higher for firms with a CL than firms without a CL; in the post-period, the odds are 74% higher for firms with a CL.

Table 3
Alignment of lawsuits and comment letters – controlling for SOX.

	(1) Pre-period 1 Jan 1997 – July 2002	(2) Pre-period 2 Aug 2002 – July 2004	(3) Pooled Pre-period 2 & Post	(4) Pooled Pre-period 2 & Post	(5) Pooled Excluding Rollout	(6) Pooled Excluding Rollout
<i>Post*CL</i>			0.357*** (0.133)	0.356*** (0.134)	0.477*** (0.144)	0.477*** (0.144)
<i>CL</i>	0.179* (0.103)	0.253** (0.127)	0.196 (0.123)	0.197 (0.124)	0.179 (0.124)	0.181 (0.124)
<i>High_volatility</i>	−0.052 (0.128)	0.768*** (0.196)	0.683*** (0.092)	0.687*** (0.092)	0.739*** (0.102)	0.745*** (0.102)
<i>Size</i>	0.026 (0.044)	−0.016 (0.070)	−0.010 (0.040)	−0.013 (0.041)	−0.016 (0.041)	−0.018 (0.042)
<i>M2B</i>	−0.007 (0.009)	0.005 (0.012)	0.006 (0.007)	0.006 (0.007)	0.009 (0.007)	0.010 (0.007)
<i>Age</i>	−0.007 (0.005)	−0.007 (0.006)	−0.011*** (0.004)	−0.012*** (0.004)	−0.011*** (0.004)	−0.012*** (0.004)
<i>Loss</i>	−0.02 (0.100)	−0.16 (0.155)	−0.042 (0.078)	−0.023 (0.078)	−0.035 (0.083)	−0.016 (0.083)
<i>Altman</i>	−0.001 (0.006)	0.017* (0.010)	0.034*** (0.005)	0.035*** (0.005)	0.036*** (0.006)	0.038*** (0.006)
<i>Chg_Sales</i>	0.214*** (0.044)	0.172* (0.089)	0.125*** (0.042)	0.125*** (0.042)	0.135*** (0.046)	0.137*** (0.046)
<i>Merger</i>	0.397* (0.218)	0.065 (0.278)	0.100 (0.094)	0.114 (0.095)	0.098 (0.099)	0.116 (0.100)
<i>Restructure</i>	0.013 (0.139)	0.025 (0.127)	−0.017 (0.074)	0.002 (0.074)	−0.034 (0.079)	−0.02 (0.080)
<i>Financing</i>	0.730*** (0.182)	0.631* (0.331)	−0.046 (0.176)	−0.064 (0.176)	−0.071 (0.191)	−0.096 (0.192)
<i>Litigation_ind</i>	−0.382** (0.154)	0.069 (0.184)	−0.18 (0.120)	−0.164 (0.120)	−0.205* (0.122)	−0.185 (0.123)
<i>Big4</i>	−0.127 (0.106)	0.229 (0.247)	−0.228** (0.114)	−0.199* (0.114)	−0.168 (0.122)	−0.134 (0.124)
<i>Tenure</i>	−0.026*** (0.009)	−0.01 (0.010)	−0.012* (0.007)	−0.012* (0.007)	−0.014** (0.007)	−0.014** (0.007)
<i>Returns</i>	−0.083 (0.054)	−0.277*** (0.093)	−0.405*** (0.064)	−0.413*** (0.065)	−0.393*** (0.068)	−0.405*** (0.069)
<i>Turnover</i>	0.563*** (0.051)	0.544*** (0.067)	0.575*** (0.044)	0.590*** (0.044)	0.566*** (0.045)	0.581*** (0.045)
<i>Skewness</i>	−0.230*** (0.049)	−0.318*** (0.076)	−0.322*** (0.035)	−0.317*** (0.035)	−0.317*** (0.039)	−0.313*** (0.039)
<i>Rstmt (t or t-1)</i>	1.711*** (0.119)	1.421*** (0.139)	0.866*** (0.092)	0.873*** (0.092)	0.863*** (0.093)	0.868*** (0.094)
<i>Amend</i>	0.195* (0.105)	0.001 (0.156)	0.311*** (0.082)	0.292*** (0.083)	0.379*** (0.092)	0.360*** (0.093)
<i>FilingGap</i>	0.303*** (0.090)	1.902*** (0.298)	1.525*** (0.124)	1.555*** (0.125)	1.593*** (0.140)	1.637*** (0.140)
<i>Constant</i>	−12.329*** (0.716)	−18.537*** (1.533)	−16.809*** (0.762)	−15.874*** (0.871)	−17.080*** (0.829)	−16.204*** (0.927)
Observations	18,911	6356	34,981	34,922	29,144	29,063
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No	Yes	No
SEC Office-Year FE	No	No	No	Yes	No	Yes
Pseudo R ²	0.210	0.214	0.174	0.184	0.174	0.185
Chi-square test of CL coefficients in (1) and (2)	χ^2 statistic: 0.21 P-value: 0.643					

This table reports the results of logistic regression $\Pr(\text{Lawsuit}) = f(\text{CL} * \text{Post}, \text{CL}, \text{controls})$ by different time periods. *CL* is an indicator variable equal to 1 if the 10 K filing in year *t* received a comment letter, and 0 otherwise. *Post* is an indicator variable equal to 1 if the time period is on or after August 1, 2004 and before June 30, 2015 and 0 otherwise. All other variables are defined in the Appendix. Column (1) includes 10-K filings between January 1, 1997 and July 31, 2002. Column (2) includes 10 K filings between August 1, 2002 and July 31, 2004. In columns (3) and (4), we pool the samples from Pre-period 2 and the post-disclosure period in column (2) of Table 2, and include year fixed effects and SEC office-year fixed effects, respectively; Columns (5) and (6) exclude the rollout period from August 1, 2004 to June 30, 2006 and include the pooled sample from Pre-period 2 and the post-period with year fixed effects and SEC office-year fixed effects, respectively. We report logistic coefficients in the table. Continuous variables are winsorized at 1% and 99% to mitigate outliers. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

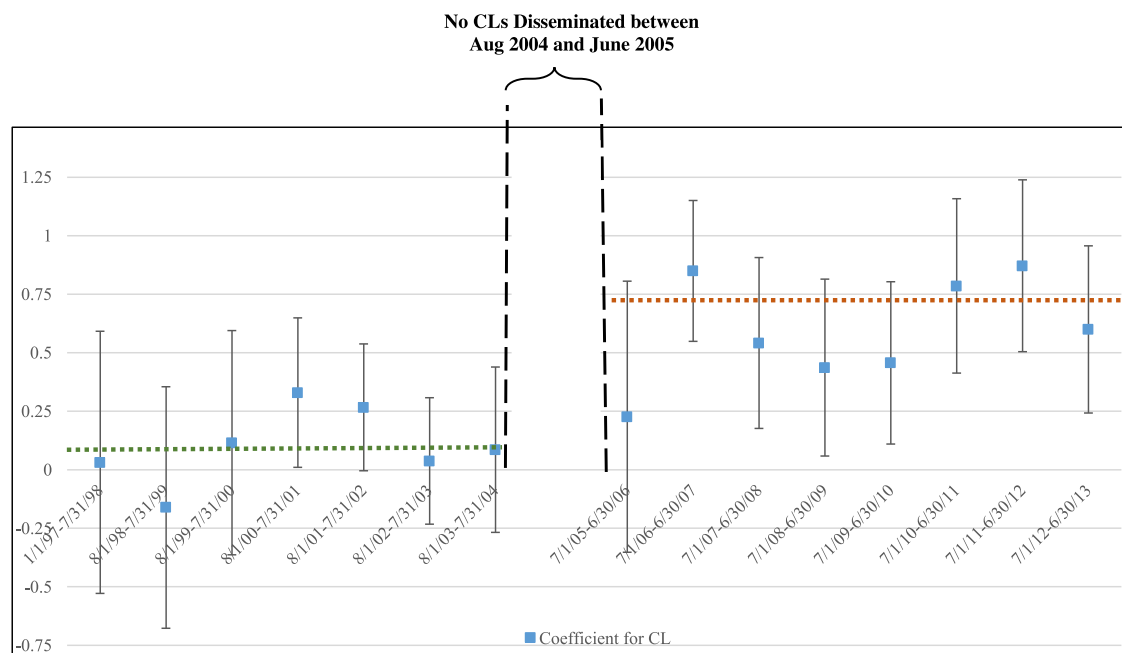


Fig. 2. Coefficients on CL (Comment Letter) by year. Fig. 2 depicts the coefficients on CL estimated using year-by-year logistic regressions: $\Pr(\text{Lawsuit}) = f(\text{CL}, \text{controls})$. CL is an indicator variable equal to 1 if the 10 K filing in year t received a comment letter, and 0 otherwise. The dashed lines represent the average of CL coefficients in the pre- and post-periods.

For firm years with 10-Ks filed after August 1, 2004 a CL is assigned to the 12-month regression period based on the dissemination date (not the 10-K filing date). Thus, there are no CLs assigned to the *No Dissemination* period (August 1, 2004 and July 1, 2005). The first estimated CL coefficient (July 1, 2005 to June 30, 2006) for the post-period falls within the Transition Period (August 1, 2004 to June 30, 2006).

Table 4 presents the regression results. We present the regression results separately for the pre-period and the post-period in columns 1 and 2, respectively. Columns 3 and 4 present the results for the pooled regressions that include the interaction term, $Rstmt * Post$.²⁵ In the pre-2004 period, the likelihood of an SEC CL is not related to a firm's impending restatements. In contrast, in the post-2004 period, an SEC CL is positively and significantly associated with eventual restatements. The chi-square test comparing the results in columns 1 and 2 rejects the equivalence of the two coefficient estimates on CL. Moreover, the coefficients on $Rstmt * Post$ in columns 3 and 4 are positive and significant, indicating that the SEC steps up its oversight of firms with questionable accounting practices after the public disclosure of its CLs.²⁶ For economic interpretation, we examine the odds ratios for coefficient estimates in column 3 (untabulated). Whereas the odds of a restating firm receiving a CL in the pre-period are about the same as the odds of a non-restating

firm receiving a CL, in the post-period, the odds of a restating firm receiving a CL are 21% higher than a non-restating firm.

To ensure that SOX Section 408 (b1) is not driving our results, we re-estimate the above regressions with the pre-disclosure period split into two sub-sample periods: Pre-period 1 (pre-SOX) and Pre-period 2 (post-SOX but pre-disclosure). We present the results in Table 5. The coefficient on $Rstmt$ is insignificant in both pre-disclosure periods. The chi-square test results confirm that the two coefficients are not significantly different from each other. We then run pooled regressions using only the post-SOX years as our pre-disclosure sample. The coefficient on the interaction term $Rstmt * Post$ is positive and significant in both columns 3 and 4.

Next, we investigate whether the SEC is less prone to political capture in the post-disclosure period. Specifically, we estimate the following logistic model to investigate the relation between PAC money and SEC CLs.

$$\text{Prob}(\text{CL}) = f(\text{PAC}, \text{PAC} * \text{Post}, \text{controls}). \quad (3)$$

PAC is the natural log of the dollar amount of a firm's PAC contribution plus one during the fiscal year.

As shown in Table 6, in the pre-2004 period, SEC CLs are significantly *negatively* associated with the PAC amount (column 1). In contrast, after 2004, the relation is reversed, as the PAC coefficient becomes positive and significant (column 2), consistent with the findings in Heese et al. (2017). The chi-square test results reject the

²⁵ Note that for the incentive-alignment tests, we do not re-estimate the regressions excluding the rollout period because the SEC's incentive alignment occurs immediately after the announcement of the policy change.

²⁶ Here, we implicitly assume that the CL is issued before a firm announces the relevant restatement, which is reasonable given the typical time lag between a wrongdoing year and the eventual restatement. We conduct an additional analysis using only restatement firms. We find that for these firms, the SEC is indeed more likely to issue a CL before the restatement announcement in the post-2004 period, compared to the pre-period.

Table 4
Incentive alignment: Comment letters and restatements.

	(1) Pre-2004	(2) Post-2004	(3) Pooled Yr FE	(4) Pooled Office-Yr FE
<i>Rstmt*Post</i>			0.143* (0.077)	0.160** (0.079)
<i>Rstmt</i>	0.048 (0.069)	0.182*** (0.039)	0.039 (0.067)	0.026 (0.069)
<i>High_volatility</i>	−0.015 (0.081)	−0.028 (0.053)	−0.012 (0.045)	−0.005 (0.046)
<i>Size</i>	0.092*** (0.021)	0.196*** (0.015)	0.157*** (0.012)	0.166*** (0.012)
<i>M2B</i>	−0.002 (0.005)	−0.004 (0.003)	−0.003 (0.003)	−0.002 (0.003)
<i>Age</i>	0.002 (0.002)	−0.002 (0.001)	0.001 (0.001)	−0.001 (0.001)
<i>Loss</i>	0.093* (0.053)	0.064* (0.037)	0.078*** (0.030)	0.086*** (0.031)
<i>Altman</i>	−0.008** (0.004)	0.000 (0.002)	−0.002 (0.002)	−0.002 (0.002)
<i>Chg_Sales</i>	−0.068* (0.039)	−0.001 (0.031)	−0.024 (0.024)	−0.018 (0.024)
<i>Merger</i>	0.145 (0.117)	−0.019 (0.037)	−0.010 (0.035)	−0.008 (0.036)
<i>Restructure</i>	−0.011 (0.060)	0.061** (0.030)	0.034 (0.027)	0.037 (0.027)
<i>Financing</i>	0.082 (0.127)	0.008 (0.093)	0.027 (0.076)	0.005 (0.076)
<i>Litigation_ind</i>	−0.011 (0.062)	−0.034 (0.039)	−0.033 (0.034)	−0.029 (0.032)
<i>Big4</i>	0.134** (0.052)	0.001 (0.037)	0.056* (0.030)	0.055* (0.030)
<i>Tenure</i>	0.001 (0.003)	−0.003 (0.002)	−0.002 (0.002)	−0.002 (0.002)
<i>Returns</i>	−0.010 (0.034)	−0.030 (0.031)	−0.035 (0.023)	−0.039* (0.023)
<i>Turnover</i>	0.117*** (0.020)	0.031** (0.012)	0.057*** (0.011)	0.053*** (0.011)
<i>Skewness</i>	0.018 (0.027)	−0.031* (0.017)	−0.011 (0.014)	−0.007 (0.015)
<i>Amend</i>	1.071*** (0.055)	0.580*** (0.050)	0.814*** (0.038)	0.803*** (0.039)
<i>FilingGap</i>	−0.673*** (0.092)	−0.259*** (0.072)	−0.429*** (0.055)	−0.443*** (0.057)
<i>Constant</i>	−1.484*** (0.470)	−1.438*** (0.380)	−2.064*** (0.306)	−1.799*** (0.496)
Observations	25,267	28,625	53,892	52,662
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No
SEC Office-Year FE	No	No	No	Yes
Pseudo R ²	0.105	0.052	0.119	0.123
Chi-square test of <i>Rstmt</i> coefficients in (1) and (2)	χ^2 statistic: 4.50 P-value: 0.034			

This table reports the results of logistic regression $\Pr(CL) = f(Rstmt*Post, Rstmt, \text{controls})$. *Rstmt* is an indicator variable equal to 1 if the current or prior firm year (*t* or *t*-1) is eventually restated, and 0 otherwise. *Post* is an indicator variable equal to 1 if the time period is on or after August 1, 2004 and before June 30, 2015 and 0 otherwise. All other variables are defined in the Appendix. Column (1) uses the sample from pre-disclosure period; Column (2) uses the sample from post-disclosure period; Columns (3) and (4) include the pooled sample from pre- and post-periods with year fixed effects and SEC office-year fixed effects, respectively. We report logistic coefficients in the table. Continuous variables are winsorized at 1% and 99%. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

equivalence of the *PAC* coefficients across the two periods. Moreover, the coefficients on *PAC * Post* are positive and significant in columns 3 and 4. Thus, our findings reconcile the evidence in [Correia \(2014\)](#) and [Heese et al. \(2017\)](#). While the SEC was likely captured when its actions were opaque to the public, it appears no longer cap-

tured after the increased transparency of its filing review process.

We repeat the above analyses in [Table 7](#) with the pre-disclosure period split into two sub-sample periods: pre-period 1 (pre-SOX and pre-disclosure) and pre-period 2 (post-SOX and pre-disclosure). The results show that

Table 5

Incentive alignment: comment letters and restatements – controlling for SOX.

	(1)	(2)	(3)	(4)
	Pre-period			
	Pre-period 1 1/1997–7/2002	Pre-period 2 8/2002–7/2004	Pooled Pre-period 2 & Post	Pooled Pre-period 2 & Post
<i>Rstmt*Post</i>			0.243** (0.096)	0.237** (0.097)
<i>Rstmt</i>	0.089 (0.099)	−0.023 (0.095)	−0.051 (0.088)	−0.046 (0.089)
<i>High_volatility</i>	−0.104 (0.102)	0.161 (0.134)	0.015 (0.050)	0.015 (0.050)
<i>Size</i>	0.083*** (0.027)	0.158*** (0.038)	0.190*** (0.014)	0.191*** (0.014)
<i>M2B</i>	−0.004 (0.006)	0.002 (0.009)	−0.003 (0.003)	−0.003 (0.003)
<i>Age</i>	0.003 (0.003)	−0.002 (0.003)	−0.002 (0.001)	−0.002 (0.001)
<i>Loss</i>	0.204*** (0.068)	−0.012 (0.088)	0.05 (0.034)	0.054 (0.034)
<i>Altman</i>	−0.011** (0.005)	−0.008 (0.007)	−0.001 (0.002)	−0.001 (0.002)
<i>Chg_Sales</i>	−0.049 (0.043)	−0.151* (0.085)	−0.018 (0.029)	−0.02 (0.029)
<i>Merger</i>	0.321* (0.179)	0.073 (0.168)	−0.017 (0.036)	−0.014 (0.036)
<i>Restructure</i>	0.075 (0.104)	−0.047 (0.077)	0.044 (0.028)	0.047* (0.028)
<i>Financing</i>	0.296** (0.147)	−0.521** (0.251)	−0.065 (0.086)	−0.071 (0.087)
<i>Litigation_ind</i>	0.004 (0.083)	0.021 (0.096)	−0.029 (0.037)	−0.027 (0.037)
<i>Big4</i>	0.089 (0.061)	0.116 (0.113)	0.001 (0.035)	0.006 (0.035)
<i>Tenure</i>	0.001 (0.005)	0.004 (0.005)	−0.002 (0.002)	−0.002 (0.002)
<i>Returns</i>	0.016 (0.043)	−0.005 (0.056)	−0.035 (0.027)	−0.033 (0.027)
<i>Turnover</i>	0.108*** (0.028)	0.099*** (0.032)	0.041*** (0.012)	0.041*** (0.012)
<i>Skewness</i>	0.041 (0.035)	−0.005 (0.045)	−0.027* (0.016)	−0.026* (0.016)
<i>Amend</i>	1.050*** (0.070)	1.141*** (0.093)	0.709*** (0.044)	0.710*** (0.045)
<i>FilingGap</i>	−0.704*** (0.110)	−0.522*** (0.179)	−0.293*** (0.067)	−0.290*** (0.067)
<i>Constant</i>	−1.221** (0.567)	−0.884 (0.890)	−1.133*** (0.351)	−1.363*** (0.499)
Observations	18,911	6356	34,981	34,981
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No
SEC Office-Year FE	No	No	No	Yes
Pseudo R ²	0.078	0.095	0.063	0.066
Chi-square test of <i>Rstmt</i> coefficients in (1) and (2)	χ^2 statistic: 0.70 P-value: 0.404			

This table reports the results of logistic regression $\Pr(CL) = f(Rstmt*Post, Rstmt, controls)$. *Rstmt* is an indicator variable and equals 1 if the current or prior firm year (*t* or *t*−1) is eventually restated, and 0 otherwise. *Post* is an indicator variable and equals 1 if the time period is on or after August 1, 2004 and before June 30, 2015 and 0 otherwise. All other variables are defined in the Appendix. Column (1) includes 10-K filings between January 1, 1997 and July 31, 2002. Column (2) includes 10K filings between August 1, 2002 and July 31, 2004. In columns (3) and (4), we pool samples from Pre-period 2 and the post-disclosure period in column (2) of Table 4, and include year fixed effects and SEC office-year fixed effects, respectively. We report logistic coefficients in the table. Continuous variables are winsorized at 1% and 99%. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

Table 6

Incentive alignment: comment letters and PAC contributions.

	(1) Pre-2004	(2) Post-2004	(3) Pooled Yr FE	(4) Pooled Office-Yr FE
<i>PAC*Post</i>			0.028*** (0.006)	0.027*** (0.007)
<i>PAC</i>	−0.023*** (0.007)	0.008** (0.004)	−0.021*** (0.006)	−0.020*** (0.006)
<i>High_volatility</i>	−0.003 (0.081)	−0.033 (0.053)	−0.010 (0.045)	−0.009 (0.045)
<i>Size</i>	0.099*** (0.022)	0.193*** (0.015)	0.162*** (0.012)	0.164*** (0.012)
<i>M2B</i>	−0.002 (0.005)	−0.004 (0.003)	−0.003 (0.003)	−0.003 (0.003)
<i>Age</i>	0.130*** (0.039)	−0.069*** (0.026)	−0.006 (0.022)	−0.008 (0.022)
<i>Loss</i>	0.105** (0.053)	0.060 (0.037)	0.079*** (0.030)	0.085*** (0.030)
<i>Altman</i>	−0.009** (0.004)	0.001 (0.002)	−0.003 (0.002)	−0.002 (0.002)
<i>Chg_Sales</i>	−0.06 (0.039)	−0.004 (0.031)	−0.025 (0.024)	−0.026 (0.024)
<i>Merger</i>	0.149 (0.117)	−0.018 (0.037)	−0.011 (0.035)	−0.008 (0.035)
<i>Restructure</i>	−0.014 (0.060)	0.063** (0.030)	0.033 (0.027)	0.036 (0.027)
<i>Financing</i>	0.086 (0.127)	0.002 (0.093)	0.028 (0.076)	0.021 (0.076)
<i>Litigation_ind</i>	−0.015 (0.062)	−0.032 (0.039)	−0.033 (0.034)	−0.033 (0.034)
<i>Big4</i>	0.131** (0.052)	−0.003 (0.037)	0.046 (0.030)	0.048 (0.030)
<i>Tenure</i>	−0.002 (0.003)	−0.002 (0.002)	−0.002 (0.002)	−0.002 (0.002)
<i>Returns</i>	−0.014 (0.034)	−0.027 (0.031)	−0.037 (0.023)	−0.036 (0.023)
<i>Turnover</i>	0.126*** (0.020)	0.027** (0.012)	0.056*** (0.011)	0.056*** (0.011)
<i>Skewness</i>	0.018 (0.027)	−0.031* (0.017)	−0.013 (0.014)	−0.012 (0.014)
<i>Rstmt (t or t-1)</i>	0.043 (0.069)	0.184*** (0.039)	0.144*** (0.034)	0.146*** (0.034)
<i>Amend</i>	1.076*** (0.055)	0.578*** (0.050)	0.816*** (0.038)	0.816*** (0.039)
<i>FilingGap</i>	−0.687*** (0.092)	−0.261*** (0.072)	−0.411*** (0.055)	−0.412*** (0.056)
<i>Constant</i>	−1.831*** (0.482)	−1.210*** (0.391)	−2.123*** (0.311)	−1.848*** (0.494)
Observations	25,267	28,625	53,892	53,848
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No
SEC Office-Year FE	No	No	No	Yes
Pseudo R^2	0.106	0.052	0.119	0.122
Chi-square test of PAC coefficients in (1) and (2)	χ^2 statistic: 13.90 P-value: <0.001			

This table reports the results of logistic regression $\Pr(CL) = f(PAC*Post, PAC, \text{controls})$. *PAC* is the natural log of annual PAC contributions in year *t*. *PAC* data are from Center for Responsive Politics (<https://www.opensecrets.org/>). *Post* is an indicator variable and equals 1 if the time period is on or after August 1, 2004 and before June 30, 2015 and 0 otherwise. All other variables are defined in the Appendix. Column (1) uses the sample from pre-disclosure period; Column (2) uses the sample from post-disclosure period; Columns (3) and (4) include the pooled sample from pre and post periods with year fixed effects and SEC office-year fixed effects, respectively. We report logistic coefficients in the table. Continuous variables are winsorized at 1% and 99%. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

the *PAC* coefficient is negative in both pre-disclosure sub-periods and significant in column 2. The chi-square test results indicate that the *PAC* coefficient in column 1 is insignificantly different from that in column 2. We pool the pre-disclosure (but post-SOX) years and post-disclosure years and include the interaction term, *PAC*Post*,

in columns 3 and 4. The coefficient on the interaction is positive and significant in both columns, suggesting that the tendency to cater to political interests tapers off significantly in the post-disclosure period, compared to the pre-disclosure (but post-SOX) period.

Table 7
Incentive alignment: comment letters and PAC – controlling for SOX.

	(1)	(2)	(3)	(4)
	Pre period			
	Pre-period 1 1/1997 – 7/2002	Pre-period 2 8/2002 – 7/2004	Pooled Pre-period 2 & Post	Pooled Pre-period 2 & Post
<i>PAC*Post</i>			0.028*** (0.009)	0.028*** (0.009)
<i>PAC</i>	−0.016 (0.010)	−0.026** (0.011)	−0.022** (0.008)	−0.022*** (0.009)
<i>High_volatility</i>	−0.101 (0.102)	0.169 (0.135)	0.015 (0.050)	0.015 (0.050)
<i>Size</i>	0.092*** (0.028)	0.175*** (0.038)	0.190*** (0.014)	0.191*** (0.014)
<i>M2B</i>	−0.004 (0.006)	0.002 (0.009)	−0.003 (0.003)	−0.003 (0.003)
<i>Age</i>	0.005* (0.003)	−0.001 (0.003)	−0.002 (0.001)	−0.002 (0.001)
<i>Loss</i>	0.209*** (0.068)	−0.007 (0.088)	0.049 (0.034)	0.053 (0.034)
<i>Altman</i>	−0.011** (0.005)	−0.010 (0.007)	−0.001 (0.002)	−0.001 (0.002)
<i>Chg_Sales</i>	−0.052 (0.043)	−0.155* (0.086)	−0.017 (0.029)	−0.019 (0.029)
<i>Merger</i>	0.323* (0.178)	0.079 (0.168)	−0.017 (0.036)	−0.014 (0.036)
<i>Restructure</i>	0.071 (0.104)	−0.048 (0.077)	0.044 (0.028)	0.048* (0.028)
<i>Financing</i>	0.288* (0.148)	−0.532** (0.251)	−0.057 (0.086)	−0.063 (0.087)
<i>Litigation_ind</i>	−0.001 (0.083)	0.01 (0.096)	−0.029 (0.037)	−0.027 (0.037)
<i>Big4</i>	0.086 (0.061)	0.102 (0.113)	−0.002 (0.035)	0.002 (0.035)
<i>Tenure</i>	0.001 (0.005)	0.004 (0.005)	−0.002 (0.002)	−0.002 (0.002)
<i>Returns</i>	0.013 (0.043)	−0.007 (0.057)	−0.036 (0.027)	−0.035 (0.027)
<i>Turnover</i>	0.111*** (0.028)	0.103*** (0.032)	0.041*** (0.012)	0.041*** (0.012)
<i>Skewness</i>	0.041 (0.035)	−0.005 (0.045)	−0.028* (0.016)	−0.027* (0.016)
<i>Restate</i>	0.087 (0.098)	−0.029 (0.095)	0.153*** (0.036)	0.154*** (0.036)
<i>Amend</i>	1.053*** (0.070)	1.150*** (0.093)	0.708*** (0.044)	0.710*** (0.045)
<i>FilingGap</i>	−0.710*** (0.110)	−0.557*** (0.180)	−0.279*** (0.066)	−0.277*** (0.067)
<i>Constant</i>	−1.269** (0.570)	−0.86 (0.891)	−1.195*** (0.351)	−1.419*** (0.493)
Observations	18,911	6,356	34,981	34,981
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No
SEC Office-Year FE	No	No	No	Yes
Pseudo R ²	0.078	0.096	0.063	0.066
Chi-square test of PAC coefficients in (1) and (2)	χ^2 statistic: 0.54 P-value: 0.463			

This table reports the results of logistic regression $\Pr(CL) = f(PAC*Post, PAC, \text{controls})$. *PAC* is the natural log of annual PAC contributions in year *t*. PAC data are from Center for Responsive Politics (<https://www.opensecrets.org/>). *Post* is an indicator variable and equals 1 if the time period is on or after August 1, 2004 and before June 30, 2015 and 0 otherwise. All other variables are defined in the Appendix. Column (1) includes 10-K filings between January 1, 1997 and July 31, 2002. Column (2) includes 10 K filings between August 1, 2002 and July 31, 2004. In columns (3) and (4), we pool the samples from Pre-period 2 and the post-disclosure period in column (2) of Table 6, and include year fixed effects and SEC office-year fixed effects, respectively. We report logistic coefficients in the table. Continuous variables are winsorized at 1% and 99%. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

4.3. Information alignment: effects of information environment

In addition to the incentive alignment, the information alignment channel can also facilitate greater complementarity between public and private enforcement after 2004. Specifically, public disclosure of SEC CLs gives private litigants access to information previously accessible only by regulators. The narrowed information gap between private litigants and the SEC likely results in a greater overlap of enforcement targets. For opaque firms with high information asymmetry, the access to SEC CLs should be particularly informative to private litigants. Thus, we expect the information channel effect to be more important for firms with high information asymmetry, such as smaller, younger firms with less institutional ownership and fewer analysts following.

We conduct cross-sectional analyses with the following four variables that capture a firm's information environment: *MktCap*, *Age*, *IO*, and *#Analyst*. We rank observations by each cross-sectional variable into ten deciles from 0 to 9. A higher decile indicates a higher value, and thus less information asymmetry. *MktCap* is the decile rank of the natural log of market capitalization. *Age* is the decile rank of the natural log of firm age. *IO* is the decile rank of the proportion of institutional investors. *#Analyst* is the decile rank of the natural log of total analysts following the firm. All cross-sectional variables are measured ex ante as the average over the pre-period for each firm. The final sample size is reduced due to missing values of the cross-sectional variables. For brevity, we suppress all control variables. Controls include *High_volatility*, *Size*, *M2B*, *Age*, *Loss*, *Altman*, *Chg_Sales*, *Merger*, *Restructure*, *Financing*, *Litigation_ind*, *Big4*, *Tenure*, *Returns*, *Turnover*, *Skewness*, *Rstmt* (*t* or *t-1*), *Amend*, *FilingGap*, *CL*, *CSVAR*, and *CL*CSVAR*.

The findings in Table 8 are generally consistent with our expectations. For example, the coefficient on *CL*Post*MktCap* is negative and significant, suggesting that the information alignment effect is more pronounced for smaller firms whose information environment is likely more opaque. Similar findings apply to younger firms and firms with lower institutional ownership. Since firm size, age, and institutional ownership all capture some aspects of a firm's information asymmetry, these findings are consistent with the information alignment effect.

4.4. Information alignment: accessibility to SEC comment letters and litigation outcomes

The improved information access by shareholder litigants also has implications for the “merit” of securities litigation. Since CL correspondence provides insight on misrepresentations and omissions in corporate filings, access to such information likely helps private shareholder litigants identify higher quality lawsuits, reducing nuisance lawsuits. We conjecture that private litigations with access to CLs are better able to target cases with merit.

To draw inference on the effect of litigants' improved information access, we need to hold the regulators' incentives constant, and we need to observe when the information is accessible to litigants. Thus, the following analysis

Table 8

Information alignment: effects of ex ante information environment.

	(1) <i>MktCap</i>	(2) <i>Age</i>	(3) <i>IO</i>	(4) <i>#Analyst</i>
<i>CSVAR</i>				
<i>CL*Post*MktCap</i>	−0.081** (0.041)			
<i>Post*MktCap</i>	−0.058** (0.027)			
<i>CL*Post*Age</i>		−0.081** (0.036)		
<i>Post*Age</i>		0.052** (0.024)		
<i>CL*Post*IO</i>			−0.088* (0.049)	
<i>Post*IO</i>			−0.042 (0.032)	
<i>CL*Post*#Analyst</i>				−0.064 (0.040)
<i>Post*#Analyst</i>				−0.066** (0.027)
<i>CL*Post</i>	0.591** (0.291)	0.380* (0.209)	0.619* (0.347)	0.474* (0.278)
<i>Post</i>	0.221 (0.260)	−0.347 (0.222)	0.167 (0.287)	0.272 (0.254)
Observations	47,718	47,718	47,718	47,718
Controls	Yes	Yes	Yes	Yes
Year FEs	No	No	No	No
Industry FEs	Yes	Yes	Yes	Yes
Pseudo R ²	0.18	0.181	0.178	0.179

This table presents the results of logistic regression $\Pr(\text{Lawsuit}) = f(\text{CL*Post*CSVAR, controls})$. All cross-sectional variables are measured as average levels over the pre-period for each firm. We then rank observations by each cross-sectional variable into 10 deciles from 0 to 9. Higher decile indicates higher value. Final sample size is reduced in this table due to missing values of measuring cross-sectional variables. *MktCap* is the decile rank of natural log of market cap. *Age* is the decile rank of the natural log of firm age. *IO* is the decile rank of the proportion of institutional shareholder. *#Analyst* is the decile rank of natural log of total analysts following the firm. We report logistic coefficients in the table. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests. For brevity, we suppress all control variables. Controls include *High_volatility*, *Size*, *M2B*, *Age*, *Loss*, *Altman*, *Chg_Sales*, *Merger*, *Restructure*, *Financing*, *Litigation_ind*, *Big4*, *Tenure*, *Returns*, *Turnover*, *Skewness*, *Rstmt* (*t* or *t-1*), *Amend*, *FilingGap*, *CL*, *CSVAR*, and *CL*CSVAR*.

is restricted to the post-disclosure period (as dissemination dates are only available for CLs issued in the post-disclosure period). We further limit the sample to only those firms with class action lawsuits that also received a CL from the SEC. Within this restricted sample, we can cleanly identify the effects of information access by exploiting the relative timing of lawsuit filing and CL dissemination. The SEC does not release CLs in real time. Instead, the full correspondence is posted to EDGAR after the review process is complete. Thus, private parties are unaware of any ongoing reviews until the SEC publicly disseminates the relevant CL correspondence.²⁷ The unit of analysis is

²⁷ We assume that the timing of a lawsuit filing is fairly exogenous. One concern is that the quality of plaintiff lawyers could be a correlated omitted variable. Specifically, the lawyers bringing pre- and post-CL cases might be of different quality, and better lawyers tend to wait until a CL is disseminated. While possible, this is not a viable strategy, as lawyers do not know which firms have received CLs. Further, the financial incen-

the individual class action lawsuit. Each lawsuit must have an associated CL so that we can compare the lawsuit's filing and CL's dissemination dates. These requirements result in a sample of 382 individual cases with CLs in the post-disclosure period.

We use the following two regressions to investigate whether litigants' access to SEC CL correspondence before filing a lawsuit is associated with two observable outcomes: whether the suit is dismissed (not certified) and the settlement amount. If the improved information access helps private litigants target cases with greater merit, then we would expect to find that these cases are less likely to be dismissed and will result in larger settlement amounts:

$$\text{Prob (Dismissed)} = f(\text{AccessBFLawsuit, controls}) \quad (4)$$

$$(\log(\text{Settlement amount})) = f(\text{AccessBFLawsuit, controls}) \quad (5)$$

As noted above, we have 382 lawsuits with CLs in the post-2004 period for the logistic regression. For the settlement regression, the lawsuit must be settled, reducing our sample size further to 136 cases. We estimate this regressing using OLS. Our variable of interest is *AccessBFLawsuit*, which equals one if the dissemination date of a CL was before the filing date of a lawsuit and zero otherwise. Because the quality of plaintiffs' representation is likely to affect the litigation outcomes, we control for lawyer quality in the regressions. We manually collect the identity of plaintiff lawyers from lawsuit complaints. We code an indicator variable, *Top10Law*, equal to one when the lead plaintiff lawyer is one of the top ten law firms (based on market share) according to the Stanford Class Action Clearing House database and zero otherwise. We include this as an additional control variable in the regressions.

We present the findings in Table 9. Once again, we include year and industry fixed effects in the regression. We find that lawsuits filed after the dissemination of the SEC CLs are less likely to be dismissed (column 1). That is, lawsuits informed by the information revealed in SEC CL correspondence are more likely to move forward in the legal proceedings and thus appear to have greater merit. The odds of dismissal for a lawsuit filed without access to CLs are more than two times that of a lawsuit filed with such access (untabulated). In addition, the settlement amounts are significantly higher for the cases filed after access to the SEC CLs (column 2).²⁸ The coefficient, 0.350, represents the increase in $\log(\text{Settlement})$ for lawsuits filed after the dissemination of the CL correspondence versus before. It translates to an increase of roughly \$3 million for the

tives to race to the courtroom suggest that plaintiff attorneys are unlikely to delay a lawsuit filing to wait for any possible CL disseminations (that may or may not materialize).

²⁸ A closely related alternative explanation is that CLs influence judges' decisions. To explore this possibility, we design a parallel test that focuses on whether the CLs are disseminated before a case is settled. If CLs affect judges' opinions, who hold sway over settlement outcomes, then we expect lawsuits with CLs available before the final settlements to have larger settlements. However, when we shift the timing of CL access from before filing to before settlement, we do not find a significant coefficient on our variable of interest.

Table 9

Information alignment: accessibility to comment letters and lawsuit outcomes.

Dependent Variable	(1) Dismiss	(2) Log_settlement
<i>AccessBFLawsuit</i>	−0.792*** (0.249)	0.350* (0.183)
<i>High_volatility</i>	0.352 (0.382)	−0.118 (0.246)
<i>Size</i>	−0.035 (0.136)	0.427*** (0.099)
<i>M2B</i>	0.079*** (0.030)	0.031 (0.024)
<i>Age</i>	−0.007 (0.009)	−0.011* (0.006)
<i>Loss</i>	−0.114 (0.326)	−0.018 (0.220)
<i>Altman</i>	0.024 (0.018)	0.018 (0.014)
<i>Chg_Sales</i>	−0.239 (0.291)	0.017 (0.119)
<i>Merger</i>	0.029 (0.324)	−0.1 (0.307)
<i>Financing</i>	−0.473 (0.724)	−0.818* (0.457)
<i>Litigation_ind</i>	0.058 (0.317)	−0.003 (0.192)
<i>Big4</i>	0.927** (0.382)	0.095 (0.260)
<i>Tenure</i>	0.003 (0.016)	−0.009 (0.014)
<i>Returns</i>	0.059 (0.247)	0.422*** (0.148)
<i>Turnover</i>	0.152 (0.146)	0.084 (0.076)
<i>Skewness</i>	0.112 (0.150)	0.064 (0.113)
<i>Rstmt (t or t-1)</i>	−0.171 (0.309)	0.518*** (0.189)
<i>Amend</i>	0.38 (0.395)	0.175 (0.220)
<i>FilingGap</i>	−0.814 (0.729)	0.16 (0.352)
<i>Top10Law</i>	−0.127 (0.247)	0.580*** (0.202)
<i>Constant</i>	−0.613 (4.001)	11.053*** (1.950)
Observations	382	136
Industry FE	Yes	Yes
Year FE	Yes	Yes
Pseudo R ²	0.120	
R		0.617

Column (1) presents the results for the logistic regression of $\text{Pr}(\text{Dismiss}) = f(\text{AccessBFLawsuit, controls})$. *Dismiss* equals 1 if a lawsuit was eventually dismissed, and 0 otherwise. *AccessBFLawsuit* equals 1 if the dissemination date of a comment letter in the post-2004 period was before the filing date of a lawsuit. Column (2) reports the results for the OLS regression of $\text{Log_settlement} = f(\text{AccessBFLawsuit, Controls})$. *Log_settlement* is the natural log of lawsuit cash settlement amount. Observations are at lawsuit level. All other variables are defined in the Appendix. The sample includes the 382 lawsuits filed in the post-disclosure period that have at least one CL issued by the SEC during the class period. In column (2) only settled lawsuits are included, reducing the sample size to 136. Size is the natural log of total assets. For these tests, our sample size is insufficient to estimate the regressions with SEC-office year fixed effects. Continuous variables are winsorized at 1% and 99%. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

median settlement.²⁹ Thus, the evidence suggests that improved information access helps private enforcement parties target cases with greater merit, resulting in fewer nuisance lawsuits.³⁰

5. Additional analyses

5.1. Alignment of private and public enforcement – timing

Having demonstrated greater alignment of enforcement targets, we next examine whether private and public enforcement activities have greater alignment in timing after 2004. In particular, we explore whether one party's enforcement action precipitates an enforcement action by the other. If so, we expect the time lag between public and private enforcement to decrease in the post-disclosure period relative to the pre-disclosure period.

We limit our sample to those firms that are subject to both private and public actions. Our unit of analysis is each class action lawsuit. We compare the 382 lawsuits with CLs filed after the 2004 disclosure policy change to the 253 lawsuits with CLs filed before the policy change. Specifically, we estimate the following OLS regressions to study whether the filing lag between the private and public enforcement actions changed from the pre- to the post-2004 period:

$$(\text{Log_AbsFilingLag}) = f(\text{Post}, \text{controls}), \quad (6)$$

where the *Log_AbsFilingLag* is the natural log of the number of days between the two enforcement actions (the issuance, not dissemination, of an SEC CL and the filing of a class action lawsuit). We present the results in Table 10. Note that the regression analysis includes all class action lawsuit observations with SEC CLs. Consistent with expectation, the coefficient on *Post* is negative and significant, suggesting that the time lag between the public and private actions has decreased after the policy change.

5.2. Falsification test

So far, we demonstrate an increase in the alignment between public and private enforcement following the 2004 policy change that resulted in public disclosure of SEC CLs. One concern is that the increased alignment is due to other factors, such as broader shifts within the SEC that altered the regulator's resources and incentives. Including SEC office-year fixed effects partially addresses this concern. To further solidify our inference that the increased alignment is attributable to the disclosure of SEC CLs, we

²⁹ To calculation in dollar value of the incremental increase in settlement, we first find the median of log (*Settlement*), 15.895, and convert it to a cash settlement: $\exp(15.895) = \$8$ million. Then, we add the coefficient estimate of 0.350 to the median log (*Settlement*), $15.895 + 0.350 = 16.245$, and convert the total to cash settlement $\exp(16.245) = \$11.35$ million. The difference is \$3.35 million.

³⁰ We conduct content analyses of the two subsets of CLs (those disseminated before and after lawsuit filings) to determine whether the content of CLs issued after lawsuit filings is systematically different from that of CLs issued before. We find no systematic differences in several dimensions, including the number of topics covered in the CLs and the types of topics raised in the CLs.

Table 10

Timing alignment: filing lag between the two enforcement actions.

Dependent Variable	(1) Log_AbsFilingLag
Post	−0.206** (0.097)
High_volatility	0.093 (0.114)
Size	−0.029 (0.046)
M2B	−0.007 (0.012)
Age	−0.001 (0.003)
Loss	0.067 (0.106)
Altman	0.001 (0.006)
Chg_Sales	−0.068 (0.067)
Merger	−0.101 (0.131)
Restructure	−0.062 (0.101)
Financing	−0.411* (0.240)
Litigation_ind	−0.069 (0.121)
Big4	−0.115 (0.110)
Tenure	−0.006 (0.008)
Returns	0.038 (0.060)
Turnover	0.080* (0.045)
Skewness	−0.056 (0.047)
Rstmt (t or t-1)	0.162 (0.100)
Amend	−0.178 (0.111)
FilingGap	0.238** (0.110)
Constant	3.912*** (0.714)
Observations	635
Industry FE	Yes
Year FE	No
R ²	0.054

This table presents the results of the OLS regression of $\text{Log_AbsFilingLag} = f(\text{Post}, \text{controls})$ using all lawsuits with comment letters. *Post* is an indicator variable and equals 1 if the time period is on or after August 1, 2004 and before June 30, 2015 and 0 otherwise. *Log_AbsFilingLag* is the natural log of the absolute value of the number of days between the first and second enforcement actions. Observations are at lawsuit level. All other variables are defined in the Appendix. The lawsuit sample includes the 382 lawsuits in the post-disclosure period with CLs and 253 lawsuits filed in the pre-disclosure period (January 1, 1997 and July 31, 2004) that also have at least one associated CL issued by the SEC during the class period. Continuous variables are winsorized at 1% and 99%. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

conduct a falsification test. We reexamine the alignment between private and public enforcers but replace the issuance of CLs with SEC investigations. SEC investigations are not publicly disclosed by the SEC, unlike SEC CLs that experienced a shift in the disclosure policy. Given the lack of transparency regarding SEC investigations throughout, we would not expect to observe increased alignment between private shareholder litigation and SEC investigations. Since the SEC does not publicly disclose which companies it is investigating or has investigated, the SEC investigations data were acquired through a FOIA request.³¹ The data set covers the period 1993 through 2018. The SEC provided a unique identifier for each investigation, the name of the targeted company, and the start and end dates of the investigation. The content of an investigation, however, is not available.

Our falsification test results are presented in Table 11. We present the results for the relation between the pre- and post-periods in columns 1 and 2, respectively. We then run a pooled regression with the interaction term, *Investigation*Post*, and present the results in columns 3 and 4. The findings consistently demonstrate no change in the association between SEC investigations and private lawsuits from the pre-period to the post-2004 period. The coefficients on *Investigation*Post* in columns 3 and 4 are not statistically significant at the 5% level. The chi-square test results fail to reject the equivalence of the coefficients on CL in columns 1 and 2. Thus, the overlap between private litigation and another form of public enforcement, SEC investigations, did not change significantly in the post-2004 period. The evidence suggests that increased alignment between shareholder litigation and SEC CLs is attributable to SEC CL disclosure.

We also plot the regression coefficients of litigation on SEC investigations and present the results in Fig. 3. While CLs experienced a shift in disclosure policy around 2004, SEC investigations have remained confidential. In contrast to the CL results, the year-by-year regressions using SEC investigations show no jump in the coefficient in the post-2004 period. The contrasting results suggest that confounding factors, such as systematic changes at the SEC or changes in lawyer quality, are unlikely to drive our main results, and therefore further solidify our inferences.

6. Robustness tests

In this section, we address various plausible alternative explanations that can confound our main inferences. First, public disclosure of internal control weaknesses arising from SOX could spuriously drive the increased association between the CL issuance and the occurrence of private litigation. Specifically, SOX Section 302 or Section 404 (requiring internal control documentation and attestation, respectively) could result in greater SEC scrutiny (hence more CLs), and disclosed internal control weaknesses could draw plaintiff lawyers' attention, leading to greater litigation risk. Under this scenario, the increased correlation

between SEC CLs and securities litigation is driven by internal control disclosures rather than the public disclosure of SEC CLs. To ensure that disclosure of internal control weaknesses is not the driver of the observed change in the correlation between CLs and litigation after 2004, we employ two separate tests. Using internal control weakness (ICW) data from Audit Analytics, we first drop all firm-years with disclosed ICWs and rerun our primary tests. Next, we include reported ICWs as an additional control variable in all our main tests.³² The tenor of the results (untabulated) is unaffected when either approach is employed.

Other potentially confounding factors include the improved quality of plaintiff representation over time or the shifting political ideology of Congressional committee members overseeing the SEC. If the quality of plaintiff representation improved after 2004, then plaintiff attorneys could better identify cases that are also of interest to the SEC, causing a spurious correlation between litigation and SEC CLs after 2004. To explore this possibility, we examine the involvement of top law firms as plaintiff attorneys before and after 2004. Presumably, these top law firms have the best-quality talent, the greatest in-house resources, and are the most experienced in securities litigation. Thus, if the leading law firms command a greater market share over time, perhaps this shift could explain our findings. This is not the case, however. The market share of the top 10 plaintiff law firms (based on the Stanford database' list of "all-time" top law firms) declined after 2004. More importantly, including the market share of top plaintiff law firms (*Big10MktShare*) as an additional control variable in our main regressions (Table 12, Panel A) does not change the tenor of our main findings.

Given that the SEC is overseen by Congress, changes in the political climate could be a correlated omitted variable. The earlier falsification test helps mitigate this concern. However, to provide further assurance, we rerun our main regressions controlling for the median political ideology of Congressional committee members overseeing the SEC. To create the needed control variable, we hand-collect data from C-Span.Org on the political ideology of committee members for the House Financial Services Committee and the Senate's Banking, Housing, and Urban Affairs. We then combine them to create a weighted average median biennial political ideology index.³³ Once we construct this index, we include it in our regressions addressing the SEC's changing incentives. The results are reported in Table 12, Panel B. Adding this additional control variable does not alter our inferences. We continue to find that the SEC is more likely to issue CLs to restating firms and firms paying higher PAC money in the post-2004 period.

³² Since ICWs are only reported after SOX took effect in August 2002, our pre-2004 sample for these additional tests includes only firm years with fiscal years ending after August 2002.

³³ The ideological positions are computed using D-NOMINATE (Dynamic Weighted Nominal Three-step Estimation procedure) developed by Poole and Rosenthal (1991). This "scaling procedure" represents legislators on a spatial map, allowing researchers to recover the "dimensions" that inform congressional voting behavior. A neutral stance or no ideological leaning is zero; right-leaning ideology is represented with positive numbers; left-leaning with negative numbers.

³¹ We obtain raw data on all closed SEC investigations between January 1, 2000 and August 2, 2017 from Blackburne et al., 2020.

Table 11
Falsification test: SEC investigations.

	(1)	(2)	(3)	(4)
	Pre-period	Post-period	Pooled Yr FE	Pooled Office-Yr FE
<i>Investigate*Post</i>			−0.196 (0.153)	−0.230 (0.153)
<i>Investigate</i>	0.576*** (0.132)	0.424*** (0.098)	0.616*** (0.130)	0.656*** (0.129)
<i>CL</i>	0.181** (0.081)	0.562*** (0.073)	0.390*** (0.055)	0.390*** (0.056)
<i>High_volatility</i>	0.193* (0.107)	0.666*** (0.104)	0.471*** (0.074)	0.467*** (0.074)
<i>Size</i>	0.028 (0.039)	−0.016 (0.047)	0.009 (0.030)	0.002 (0.030)
<i>M2B</i>	−0.004 (0.008)	0.007 (0.008)	0.001 (0.005)	0.001 (0.005)
<i>Age</i>	−0.008** (0.004)	−0.013*** (0.004)	−0.011*** (0.003)	−0.012*** (0.003)
<i>Loss</i>	−0.035 (0.086)	−0.033 (0.090)	−0.029 (0.062)	0.004 (0.062)
<i>Altman</i>	0.001 (0.005)	0.040*** (0.006)	0.017*** (0.004)	0.020*** (0.004)
<i>Chg_Sales</i>	0.212*** (0.038)	0.128*** (0.046)	0.177*** (0.029)	0.181*** (0.030)
<i>Merger</i>	0.229 (0.186)	0.117 (0.098)	0.078 (0.090)	0.084 (0.091)
<i>Restructure</i>	−0.018 (0.098)	−0.054 (0.085)	−0.057 (0.066)	−0.049 (0.066)
<i>Financing</i>	0.716*** (0.161)	−0.188 (0.195)	0.295** (0.128)	0.282** (0.129)
<i>Litigation_ind</i>	−0.253* (0.134)	−0.290** (0.137)	−0.250** (0.099)	−0.235** (0.098)
<i>Big4</i>	−0.12 (0.098)	−0.278** (0.123)	−0.251*** (0.076)	−0.205*** (0.077)
<i>Tenure</i>	−0.019** (0.007)	−0.011 (0.008)	−0.016*** (0.005)	−0.016*** (0.005)
<i>Returns</i>	−0.130*** (0.048)	−0.488*** (0.083)	−0.256*** (0.042)	−0.256*** (0.043)
<i>Turnover</i>	0.530*** (0.044)	0.578*** (0.053)	0.533*** (0.034)	0.559*** (0.035)
<i>Skewness</i>	−0.251*** (0.042)	−0.325*** (0.039)	−0.303*** (0.028)	−0.298*** (0.028)
<i>Rstmt (t or t-1)</i>	1.571*** (0.100)	0.673*** (0.108)	1.141*** (0.075)	1.154*** (0.075)
<i>Amend</i>	0.159* (0.087)	0.419*** (0.093)	0.273*** (0.064)	0.260*** (0.064)
<i>FilingGap</i>	0.512*** (0.085)	1.487*** (0.134)	0.870*** (0.074)	0.886*** (0.073)
<i>Constant</i>	−12.906*** (0.648)	−16.762*** (0.856)	−14.459*** (0.569)	−14.383*** (0.912)
Observations	25,267	28,625	53,892	53,605
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No
SEC Office-Year FE	No	No	No	Yes
Pseudo R ²	0.206	0.170	0.175	0.186
Chi-square test of <i>Investigate</i> coefficients in (1) and (2)	χ^2 statistic: 0.93 P-value: 0.335			

This table reports the results of logistic regression $\Pr(\text{Lawsuit}) = f(\text{Investigate*Post}, \text{Investigate}, \text{controls})$. *Investigate* is an indicator variable equals 1 if a firm in year *t* was investigated by the SEC, and 0 otherwise. All other variables are defined in the Appendix. Column (1) uses the sample from pre-disclosure period; Column (2) uses the sample from post-disclosure period; Columns (3) and (4) include the pooled sample from pre- and post-periods with year fixed effects and SEC office-year fixed effects, respectively. We report logistic coefficients in the table. Continuous variables are winsorized at 1% and 99%. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

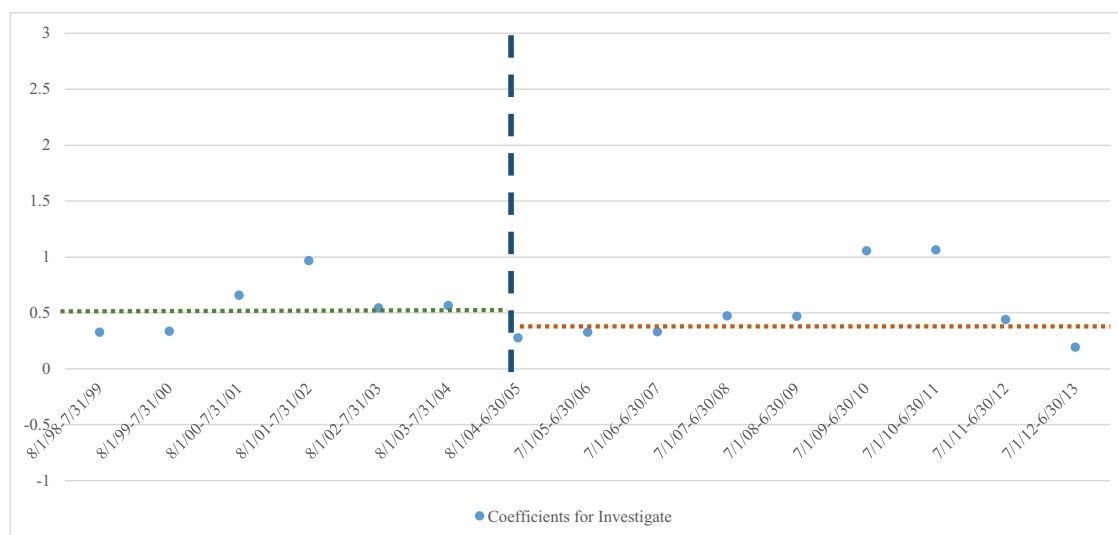


Fig. 3. Coefficients on *Investigate* (SEC Investigation) by year – falsification test. Fig. 3 depicts the coefficients on SEC *Investigate* estimated using year-by-year logistic regressions: $\Pr(\text{Lawsuit}) = f(\text{Investigate}, \text{controls})$. The horizontal dashed lines represent the average of *Investigate* coefficients in the pre- and post-periods. The vertical dashed line represents the public disclosure of SEC comment letters in August 2004.

Another possible confounding factor is the content of the CLs, which could have changed around 2004 due to the evolving SOX requirements or changes in the SEC's general approach in reviewing registrants' filings unrelated to the policy change. To address this concern, we obtained a sample of 168 pre-2004 (useable but incomplete) CL conversations between the SEC and the target firms.³⁴ We compare the content of these pre-disclosure CLs to that of publicly disclosed CLs covered in the Audit Analytics database. If CLs' content evolved in the post-2004 period in a manner likely to draw more attention from litigation attorneys, we would expect the post-2004 CLs to cover more topics, discuss more severe issues, and contain stronger language.

Contrary to this, we find that the average number of issues or topics raised in CLs was significantly higher in the pre- (vs. the post-2004) disclosure period. The categories of topics discussed more frequently in the pre-2004 CLs include the following: Accounting, Disclosure, Auditor Related, Legal Matters, and Restatements (including accounting errors). The only category discussed

more frequently after 2004 than before 2004 was Internal Control issues, which likely resulted from SOX Section 404. In addition, "strong language" is used more frequently in the pre-2004 CLs, as are discussions of DOE involvement (including discussion of SEC investigations that were not disclosed to the public). Based on detailed, comparative textual analysis, it appears that the SEC softened the language used in its CLs and halted discussions of DOE investigations once the CL correspondence became public. Thus, if there is any effect, it would make lawsuits less likely to be correlated with CLs after 2004 because the post-2004 CLs use strong language less frequently and do not discuss the involvement of the DOE.

7. Conclusion

We find that enhanced regulatory transparency facilitates the alignment between private and public enforcement actions. We study the interaction between a public enforcement entity (the SEC) and a private enforcer mechanism (private securities litigation) before and after an SEC policy change that resulted in the SEC publicly disclosing its comment letter correspondence with issuers. We find that after the public disclosure of comment letters, the SEC and the private litigants achieved greater alignment in their respective enforcement targets. Firms that receive SEC comment letters are also more likely to be targets of private securities litigation compared to the pre-disclosure period. Importantly, our empirical work indicates that the increase in the overlap between comment letters and class actions is not driven by the SOX Section 408, which requires that the SEC review issuers' filings at least once every three years. Instead, only after the SEC changed its disclosure policy in August 2004 did the overlap increased significantly.

³⁴ We obtained this sample via earlier FOIA requests (filed by several colleagues and us). To ensure consistency between our coding (for the pre-2004 period) and the coding done by Audit Analytics (for the post-2004 comparison), we hired the same person that oversaw the coding of the publicly disclosed CLs (post-2004) for Audit Analytics. We code the following characteristics from each CL: total # of topics raised; # of accounting topics; # of disclosure-related topics; whether an auditor is mentioned in the CL; whether a CL discussed internal control issues, legal matters, risk factors, 8-K disclosures or accounting restatements. In addition, we manually searched the 168 pre-2004 CLs and coded those with "strong language" (search string: "we do not agree" OR "disagree" OR "inadequate") and those discussing the involvement of the Division of Enforcement (search string: "Enforcement" OR "Investiga"). We used the same search strings for the post-2004 public CLs available on Audit Analytics.

Table 12
Robustness checks.

	(1) Pooled	(2) Pooled Pre-period 2 & Post		
Panel A: Alignment of lawsuits and comment letters – controlling for quality of law firms				
<i>CL*Post</i>	0.198** (0.100)	0.420*** (0.132)		
<i>CL</i>	0.279*** (0.078)	0.156 (0.119)		
<i>Big10MktShare</i>	0.693*** (0.138)	1.253** (0.518)		
Observations	53,892	34,981		
Controls	Yes	Yes		
Industry FE	Yes	Yes		
Year FE	No	No		
SEC Office-Year FE	No	No		
Pseudo <i>R</i> ²	0.167	0.171		
	Variable is <i>Rstmt</i>		Variable is <i>PAC</i>	
	(1) Pooled	(2) Pooled Pre-period 2 & Post	(3) Pooled	(4) Pooled Pre-period 2 & Post
Panel B: Incentive alignment – controlling for politician ideology				
<i>Post*Variable</i>	0.457*** (0.073)	0.325*** (0.090)	0.063*** (0.006)	0.042*** (0.008)
<i>Variable</i>	−0.102 (0.064)	−0.129 (0.084)	−0.052*** (0.006)	−0.035*** (0.008)
<i>Ideology</i>	−2.270*** (0.074)	−1.562*** (0.073)	−2.239*** (0.074)	−1.561*** (0.073)
Observations	53,892	34,981	53,892	34,981
Controls	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	No	No	No	No
SEC Office-Year FE	No	No	No	No
Pseudo <i>R</i> ²	0.097	0.055	0.098	0.055

Panel A presents the results of logistic regression $\Pr(\text{Lawsuit}) = f(\text{CL*Post}, \text{CL}, \text{controls})$. *CL* is an indicator variable equal to 1 if the 10 K filing in year *t* received a comment letter, and 0 otherwise. *Post* is an indicator variable equal to 1 if the time period is on or after August 1, 2004 and before June 30, 2015 and 0 otherwise. We include *Big10MktShare* as a control. *Big10MktShare* is an annual index representing the proportion of all 10b-5 lawsuits that involved at least one of the top 10 law firms. Top 10 law firms are defined by Stanford Class Action Clearinghouse (http://securities.stanford.edu/top-ten.html?filter=plaintiff_firm). They are Milberg LLP, Robbins Geller Rudman & Dowd LLP, Topaz Kessler Meltzer & Check LLP, Wolf Haldenstein, Stull & Brody, Bernstein Liebhard LLP, Pomerantz LLP, Sirota & Sirota, Bernstein Litowitz Berger & Grossmann LLP, and Labaton Sucharow. Columns (1) includes the pooled sample from pre- and post-periods, respectively. In columns (2), we pool the samples from Pre-period 2 (August 1, 2002 and July 31, 2004) and the post-disclosure period. Panel B reports the results of logistic regressions $\Pr(\text{CL}) = f(\text{Post} * \text{Variable}, \text{Variable}, \text{controls})$. *Variable* is *Rstmt* in columns (1) and (2), and *PAC* in columns (3) and (4). Columns (1) and (3) include the pooled sample from pre- and post-periods. In columns (2) and (4), we pool the samples from Pre-period 2 (August 1, 2002 and July 31, 2004) and the post-disclosure period. We suppress all control variables for brevity. Controls include *High_volatility*, *Size*, *M2B*, *Age*, *Loss*, *Altman*, *Chg_Sales*, *Merger*, *Restructure*, *Financing*, *Litigation_ind*, *Big4*, *Tenure*, *Returns*, *Turnover*, *Skewness*, *Rstmt* (*t* or *t-1*), *Amend*, and *FilingCap*. All variables are defined in the Appendix. We report logistic coefficients in the table. Continuous variables are winsorized at 1% and 99%. Standard errors are clustered at the firm level and reported in parentheses. *, **, and *** denote significance at the 0.1, 0.05, and 0.01 levels, respectively, using two-tailed tests.

We hypothesize and find that the increased alignment between public and private enforcement is attributable to two channels. First, the increased visibility of the SEC's actions enhances the SEC's incentives and reduces regulatory capture. We find that the SEC is more likely to issue comment letters to firms with questionable accounting practices (i.e., firms that eventually must restate their financial reports). This finding suggests that the SEC increased its oversight in the post-disclosure period. We also find that firms making higher PAC contributions in the post-disclosure period are more likely to receive SEC comment letters, consistent with Heese et al. (2017). This evidence is in sharp contrast to the SEC's behavior in the pre-disclosure period, when the SEC was less likely to issue comment letters to firms with high PAC contributions.

Collectively, the results suggest that the SEC steps up its enforcement efforts and becomes less sensitive to political pressure once its review process is more observable to the public. These findings should be of interest to regulators, including Congress (the regulator of regulators), as they suggest that greater transparency of regulators' oversight activities leads to greater regulatory effectiveness and independence.

Second, public disclosure of SEC comment letters enhances private litigants' access to information, which in turn helps litigants identify and pursue high-quality cases, limiting nuisance cases. We show that when plaintiffs have access to the comment letter correspondence before they file their cases, the lawsuits have greater merit as the dismissal rate is lower for these cases than for

lawsuits filed without access to the relevant SEC correspondence. Moreover, settlement amounts are also larger for lawsuits filed with the knowledge of SEC comment letters.

Overall, we contribute to the broad academic debate on the effectiveness of various enforcement actions by providing evidence that enhanced regulatory transparency facilitates greater alignment of public and private enforcement actions, via improved incentives and enhanced infor-

mation access. Greater overlap demonstrates the complementarity of public and private enforcement. Private litigants, aided by greater regulatory transparency, appear to pick up where the public enforcement stops (due to resource constraints or incentive problems). Ultimately, the alignment of the two mechanisms leads to more efficient enforcement if the engagement of both holds more corporate wrongdoers accountable while reducing deadweight loss associated with nuisance cases.

Appendix: Variable definitions

Variable Name	Definition
<i>Post</i>	An indicator variable equal to 1 if the time period is on or after August 1, 2004 and before June 30, 2015 and 0 if the time period is between January 1, 1997 and July 31, 2004.
<i>CL</i>	An indicator variable equal to 1 if the 10-K filing in year <i>t</i> received a comment letter, and 0 otherwise.
<i>PAC</i>	Natural log of annual PAC contributions plus one. The data are from Center for Responsive Politics (https://www.opensecrets.org/).
<i>Rstmt (t or t-1)</i>	An indicator variable equal to 1 if the current or prior firm-year is eventually restated, and 0 otherwise. Restatement information is from GAO restatement database and Audit Analytics.
<i>High_volatility</i>	An indicator variable equal to 1 if a firm's stock volatility ranks top 10% in its industry year, and 0 otherwise.
<i>Size</i>	Natural log of total market capitalization.
<i>M2B</i>	Market-to-book ratio is calculated as total market capitalization divided by book value.
<i>Age</i>	Firm age is the number of annual observations in Compustat through year <i>t</i> .
<i>Loss</i>	An indicator variable equal to 1 if earnings before extraordinary items is negative in a given year, and 0 otherwise.
<i>Altman</i>	Altman <i>z</i> score is calculated based on Altman (1968) and Defond and Hung (2003) and is equal to $1.2 \times (\text{net working capital (ACT-LCT)}/\text{total assets (AT)}) + 1.4 \times (\text{retained earnings (RE)}/\text{total assets}) + 3.3 \times (\text{earnings before interest and taxes (PI - XINT)}/\text{total assets}) + 0.6 \times (\text{market value of equity (CSHO} \times \text{PRCC_F)}/\text{book value of liabilities (LT)}) + 1.0 \times (\text{sales (SALE)}/\text{total assets})$.
<i>Chg_Sales</i>	Percentage change in sales revenue (REVT in Compustat) from year <i>t</i> -1 to year <i>t</i> .
<i>Merger</i>	An indicator variable equal to 1 for non-zero acquisitions or mergers as reported on a pre-tax basis (AQP) in Compustat in year <i>t</i> and 0 otherwise.
<i>Restructure</i>	An indicator variable equal to 1 for non-zero restructuring costs as reported in Compustat on a pre-tax basis (RCP) in year <i>t</i> and 0 otherwise.
<i>Financing</i>	The sum of equity financing and debt financing scaled by total assets, measured in <i>t</i> + 1, following Ettredge et al., (2011). Equity financing equals the sales of common and preferred stock (SSTK) minus the purchases of common and preferred stock (PRSTKC) minus dividends (DV). Debt financing equals long-term debt issued (DLTIS) minus long-term debt reduction (DLTR) minus the change in current debt (DLCCH).
<i>Litigation_ind</i>	An indicator variable equal to 1 if the company is in a highly litigious industry (four-digit SIC industry codes 2833–2836, 3570–3577, 3600–3674, 5200–5961, 7370–7374, or 8731–8734 following Francis et al. (1994) and Kim and Skinner (2012), and 0 otherwise.
<i>Big4</i>	An indicator variable equal to 1 if the auditor is a Big 4 audit firm, and 0 otherwise.
<i>Tenure</i>	Auditor tenure is the number of years during which the auditor has audited the company.
<i>Returns</i>	Market-adjusted 12-month stock return for year <i>t</i> -1.
<i>Turnover</i>	Natural log of trading volume accumulated over the 12-month period in year <i>t</i> -1 ending with the fiscal year-end scaled by beginning of year <i>t</i> -1 shares outstanding, plus one.
<i>Skewness</i>	Skewness of the firm's 12-month return for year <i>t</i> -1.
<i>Amend</i>	An indicator variable equal to 1 if the firm filed an amended annual filing, and 0 otherwise.
<i>FilingGap</i>	Natural log of the number of days between fiscal year end date and 10-K filing date plus one.
<i>AccessBFLawsuit</i>	An indicator variable equal to one if the dissemination date of comment letter in the post-2004 period was before the filing date of lawsuits, and 0 otherwise.
<i>Dismiss</i>	An indicator variable equal to one if the lawsuit is eventually dismissed, and 0 otherwise.
<i>Log_settlement</i>	Natural log of the lawsuit cash settlement amount.
<i>Log_AbsFilingLag</i>	Natural log of the absolute value of the number of days between the first and second enforcement actions.
<i>Top10Law</i>	An indicator variable equal to one if the lead plaintiff's lawyer is one of the top 10 law firms (including their predecessors and successors), and 0 otherwise. Top 10 law firms are defined by Stanford Class Action Clearinghouse as the 10 firms with the largest market shares in securities litigations (http://securities.stanford.edu/top-ten.html?filter=plaintiff_firm). They are Milberg LLP, Robbins Geller Rudman & Dowd LLP, Topaz Kessler Meltzer & Check LLP, Wolf Haldenstein, Stull & Brody, Bernstein Liebhart LLP, Pomerantz LLP, Sirota & Sirota, Bernstein Litowitz Berger & Grossmann LLP, and Labaton Sucharow.
<i>Investigate</i>	An indicator variable equal to 1 if the firm-year was investigated by the SEC, and 0 otherwise.

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